SYNSPEC2.01: User's guide

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Abstract

We present a users's manual for newly upgraded program SYNSPEC, version 2.01, together with a number of utility programs for constructing SYNSPEC-compatible molecular line lists from the EXOMOL data, and a program CHEMEQ for a fast solution of chemical equilibrium (the equation of state.

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1 Introduction

This document represents a detailed technical description and a user's manual for working with the spectrum synthesis program SYNSPEC, described in general terms in an accompanied paper (Hubeny, Allende Prieto, & Osorio, 2026; hereafter referred to as Paper V).

Besides the SYNSPEC itself, we also present a number of utility programs, such as the suite of routines to generate molecular line lists in the SYNSPEC format from EXOMOL data, and a short program CHEMEQ for a fast exploration of a solution of chemical equilibrium under various conditions, and with various molecular data. This program is also supplemented by a Python program to display the results in a graphical way.

Finally, we will briefly describe two wrappers around SYNSPEC, which generally prepare input data, run SYNSPEC, and graphically display the results. One is an older IDL program SYNPLOT, which still exists but is not maintained any more because of a limited availability of IDL, and a more recent one is the Python code SYNPLE.

2 General outline

2.1 Background

The physical background of SYNSPEC was discussed in some detail in Paper V. Briefly, SYNSPEC solves, frequency by frequency, the radiative transfer equation for a 1-dimensional slab (stellar atmosphere, or a segment of an accretion disk), with the depth dependence of physical parameters (temperature, density, etc.) specified by the input model atmosphere. The end result is the emergent radiation flux (or intensity) at the surface of the atmosphere, which may then compared to an observed spectrum, with the aim to determine as many physical properties of a studied object as possible.

The user can select a degree of complexity of treating atomic and molecular opacity sources, lines and continua. Very few necessary atomic and molecular parameters are hardwired in the program. Instead, SYNSPEC is distributed together with a significant amount of such data, and the user may select which actual data are being used, and for which atomic and molecular species. Such a selection is accomplished by setting relevant input keyword parameters that are described in detail in this document. Each keyword is assigned a default value, so if no keywords are specified, SYNSPEC still performs some reasonable calculations. Nevertheless, several keyword parameters are crucial, such as a switch to select the basic mode of operation, setting the wavelength range, to turn on/off molecules, and a number of molecular line lists, etc.

Besides computing a synthetic spectrum of a star or an accretion disk, possibly also a giant planet, SYNSPEC offers several other applications, such as computing an opacity table that can be used for a model atmosphere construction by the program TLUSTY (Hubeny & Lanz 1995). Detailed user's guide and

manual for this program is presented in the series of papers by Hubeny & Lanz (2017 a,b,c - hereafter referred to as Paper I, II, and III); and Hubeny et al. (2021; Paper IV). These papers also contain a description and users' manual for the previous versions of SYNSPEC, namely for version 51, now called 1.51 (Paper I), and version 54 (1.54) in Paper IV. Although the content and meaning of many input files and keyword parameters remains unchanged with respect to the previous versions, there are still significant changes and improvements in the present one. Therefore, we describe in this Manual all the input files and parameters, regardless whether they were already described in the earlier papers.

2.2 Programming changes

As already mentioned in Paper V, the most significant coding improvement of the current version 2.01 of SYNSPEC is that was transformed to FORTRAN90/95. Strictly speaking, not all available features of the language were used, so here is the list of what changes that have been made:

• Most importantly, all COMMON blocks were removed and replaced by MODULES.

• Most arrays were allocated, instead of specifying their dimensions in advance. The allocations take into account the nature and extent of intended calculations.

• Arrays related to line lists and line atomic data are allocated dynamically, taking into account actual numbers of lines to be dealt with. In particular, this allows the program to work with an arbitrary number of individual molecular line lists.

• All labels were removed and replaced by specific constructs, such as `CYCLE, EXIT` (from loops), `SELECT CASE`, etc., and format strings.

• Instead of `REAL*8`, we use more modern specification `REAL(DP)`, with `DP` defined through an appropriate `DP=SELECTED_REAL_KIND(15,300)`.

• In a few subroutines the recommended `IMPLICIT NONE` statement was used, but not in all. Since the code was written strictly adhering to the implicit definition of integer variables as those staring with I-N, such an additional effort of naming explicitly all variables in all subroutines was deemed as an unnecessary cosmetic change. But it certainly can be done in future releases.

• Comment lines starting with `c` or `C` were replaced by `!`, and continuation lines were identified the character "&" in the 6th position, and in the previous line by the character "&" in the 75th position. In this way, the program is compatible with both fixed format and free format, and the source files may have extension either `*.f`, or `*.f90`(`95`), and will be treated by a compiler in its default setting without the necessity to specify the format.

• Traditional relation operators such as `lt`, `gt`, etc., were kept, instead of replacing them with "<, >", etc.

• Shortcuts of dealing with arrays as first-class objects were used only very sporadically.

The source code is now distributed as 3 files, instead of previous 7 files, namely:

• File `synspec2.01.f`, a collection of all executive routines.

• File `syn_modules.f`, a collection of all `modules`.

• File `syn_alloc.f`, a collection of all allocation subroutines.

The last two files are kept separately, in order to allow an easy modification of modules, and/or allocations of arrays.

3 Basic modes of operation of SYNSPEC

In this section, we describe the operation in the general way, without necessarily specifying the parameters that switch on the individual options. We refer to the individual parameters as *keyword parameters*. In §§ 4.2 and 4.3 we explain how are these parameters specified and used.

Program SYNSPEC can be used for a variety of purposes. The amount of input, specifically the number and a content of input files, may be different for different applications of SYNSPEC.

There are in principle four types of applications: (i) computing a stellar synthetic spectrum, which is the original and standard application; (ii) analogously, computing a spectrum for one segment of an accretion disk; (iii) constructing an opacity table for a further use by TLUSTY, and (iv) other purposes, such as a line identification table, the so-called iron-curtain option, or just a solution of the equation of state. The choice of the mode of application is set by the input keyword parameter `IMODE`, which will be described in the next section. We summarize the applications and the corresponding input files in turn.

3.1 Constructing a synthetic spectrum of a star

There are three possibilities:

- IMODE=0 – the *standard mode*. In the case SYNSPEC computes a synthetic spectrum, including lines referred to in the list list(s), and specified as described below. Other quantities, such as the wavelength range at which the spectrum is computed, line strength cutoff, etc., are specified through the appropriate keyword parameters.

- IMODE=1 – a *few lines mode*. This is an outdated option still kept for downward compatibility. It is analogous to the previous one, but the line list may contain just one or very few lines. It was originally introduced to facilitate studying a behavior of a single line or a multiplet (e.g. the Mg II resonance doublet), without a "pollution" from overlapping weak lines. But during the lifetime of SYNSPEC, it was rarely used.

- IMODE=2 – a *continuum-only mode*. As the name suggests, all spectral lines of elements other than hydrogen (and optionally also ionized helium) are disregarded. SYNSPEC considers these lines a part of the "continuum", in the sense that their physical properties (wavelengths, oscillator strengths, etc.) are hardwired in the program. These lines do not appear in the line list (and if they do. an information from the line list is disregarded). From the point of view of astronomical applications, the option is useful because the hydrogen lines may be very wide, and indeed provide a sort of "continuum" for weaker overlapping metal lines.

In all these cases, the following input files are mandatory: The first category of such files are those which would be used for TLUSTY, so they are provided from the source of TLUSTY models. These are:

• Standard input file – main control data, specifying the chemical composition, a selection of atomic energy levels treated explicitly (and consequently the bound-free transitions from them being taken into account), and a specification of the free-free transitions. This file may also contain atomic data for lines that were taken into account for model atmosphere construction, but these are disregarded by SYNSPEC because it takes information about lines from different sources (line lists). If a model was not constructed by TLUSTY, this standard input file is set up to simulate the TLUSTY input, i.e, with the appropriate chemical composition, and bound-free and free free cross-sections. An example of such file is presented in § 4.1.

• `fort.8` – Input model atmosphere - typically given as a previously computed TLUSTY model, but it can also be a Kurucz model. It can also be a model from other sources, such as the MARCS models (Gustafsson er al. 1994), but in this case the user has to transform the model file to the TLUSTY format. This is actally done in the Python wrapper SYNPLE – see § 6.2.

• – For options `IMODE=0` or 1, the line list(s) is(are) necessary. There is a single atomic line list, and, if molecules are present, there is one or more molecular line lists. Since line lists are required for several other options, they will be described in more detail in § 4.7.

3.2 Constructing a synthetic spectrum for an accretion disk

In this case SYNSPEC computes a synthetic spectrum for one segment (ring) of an accretion disk, and the total spectrum of the disk is obtained by integrating synthetic spectra for all individual rings (e.g., Linnell & Hubeny 1996; Hubeny & Long 2021). From the point of view of spectrum synthesis, a single ring behaves exactly like a stellar atmosphere. Therefore, computing its spectrum is completely analogous to the stellar case (the only small difference being the inner boundary condition for the radiative transfer equation).

Computationally, the three options of computing the spectrum (`IMODE`=0,1,2), are still in effect, and the disk option is switched on by setting the keyword parameter `INMOD`= 2.

All potential additional files, such as line lists, etc., are exactly the same for the "disk" and the "stellar" mode.

3.3 Constructing an opacity table

This mode is invoked by setting `IMODE` $= -3$ or -4. In this case, the opacity is computed exactly the same way as in the previous modes. However, there is no input model atmosphere; instead the input file `fort.2` (see § 4.8) specifies a set of temperature/density, (T, ρ), pairs, and one such pair is formally considered an "atmosphere" with just one depth point. The opacity is computed in the same way as for a normal atmosphere. SYNSPEC then loops over all (T, ρ) pairs, and stores the opacities in a specific output file. There are two options:

- `IMODE=-3` – regular opacity table with lines;
- `IMODE=-4` – "continuum-only" opacity table; analogous to the mode `IMODE=2` for a regular spectrum synthesis.

Since this application of SYNSPEC is just a variant of the "normal" spectrum synthesis, with the only difference being not solving the transfer equation. All additional input files, besides the input mode atmosphere part, have to be present.

3.4 Other applications

The two following options are more of less outdated, but are kept for the sake of downward compatibility:

- `IMODE=-2` *line identification* option. When it is switched on, the program performs a selection of lines that are expected to contribute to the opacity by computing the line-center and the continuum opacities in one depth point, called *standard depth* – see Appendix A. The program does not solve the transfer equation to get specific intensities, but instead computes approximate equivalent widths of lines. Its output is `fort.12`, an identification table, that shows approximate strength of lines. It is therefore very useful for identification of lines in the synthetic spectrum.

  The same file is produced in the standard run of SYNSPEC (with `IMODE`=0 and `IPRIN` ≥ 1), but if the file is not produced or deleted by mistake, or if the user just wants to get approximate line strengths for a given model atmosphere, this option is useful because it avoids more time-consuming (re-)computation of the actual spectrum.

- `IMODE=-1` the *iron curtain* mode. It is outdated because it is equivalent to producing an opacity table with just one single (T, ρ) point.

- `IFEOS`> 0, with `IFEOS` being one of the keyword parameters. In this mode, SYNSPEC merely generates an output with several tables showing the results of the chemical equilibrium (EOS) calculations, without computing opacities and the spectrum.

However, this option does very much the same as the auxiliary program CHEMEQ, described in Appendix D. The difference between these two approaches is that the option `IFEOS`> 0 produces the EOS tables for several chosen depths in the actual atmosphere, while the program CHEMEQ produces such tables for a single chosen (T, ρ) pair, but with a more comprehensive output.

3.5 Setting the internal wavelengths

Although the procedure of setting the internal wavelength points is automatic, and completely transparent to the user, we will briefly describe it here because it explains the usage of some keyword parameters, and clarifies the meaning of some terms used in this document.

The setting of inner wavelength points is different when one deals with full opacity including lines [modes `IMODE` $= 0$ (spectrum) or -3 (opacity table)], and the continuum-only case (`IMODE` $= 2$ or -4). The latter case is simple because the wavelength points are logarithmically equidistant between the starting and ending wavelength, specified by the keyword parameters `ALAMO` and `ALAST` – see § 4.3.1. The spacing of the wavelength points is set by the keyword `SPACE`.

In the standard case where the lines are being considered, the procedure is as follows. The programs first goes through the atomic line list, and reject lines that are too weak to contribute, using the procedure described in Appendix A.2. The lines are also rejected if their central wavelengths are outside of the wavelength range (`ALAMO-CUTOFF, ALAST+CUTOFF`); with `CUTOFF` being one of the keyword parameters that specifies the distance in the wavelength space to which a line can in principle contribute to the opacity. This allows for a contribution of lines whose centers are out of the range of computed spectrum, but which can still potentially contribute through their wings. These keywords are described in § 4.3.1. The parameters for the lines that are selected are stored in appropriate arrays. The analogous procedure is repeated for every molecular line list. The maximum number of selected lines from an atomic line list is called `MLINO`, and from any molecular line list as `MLINMO`. The default value of both parameters is 10^6, but this can be easily changed by setting an input file called `dimens`, as explained in § 4.9.

The program then sets the first wavelength at `ALAMO`, and finds the first selected line with a larger central wavelength, λ_0. It than sets two wavelength points, one at the midpoint between the previous one, and one at λ_0. If the interval between points is larger than the value specified by the keyword parameter `SPACE`, the program sets more wavelength points so that the distance between points is not larger than `SPACE`. The program then goes to the next selected line, and repeats the procedure. In this way it is assured that no line center and no in-between space (continuum window) is missed. Once the number of so determined wavelength points reaches a limit, set up by the keyword parameter `NSFREQ`, the procedure is terminated, and these points form the so-called "set". The first and last wavelength of the "set" are the points where the continuum opacity is evaluated exactly, while for all the internal points it is interpolated.

As will be described later on, the NSFREQ keyword has the default value 120, and the SPACE keyword 0.01, so the typical length of a "set" is about 1 Å, and therefore the continuum opacity is evaluated at a reasonably dense mesh of wavelengths. Choosing NSFREQ too large would lead to a large distance between the points at which the continuum opacity is evaluated exactly, and thus to potential inaccuracies.

We stress that since the overall spectral range between ALAMO and ALAST may be quite large, the input parameters CUTOFF and SPACE are defined for a midpoint, $\lambda_c = (\texttt{ALAMO} + \texttt{ALAST})/2$, and for each "set" they are scaled by λ_0/λ_c, where λ_0 is the initial wavelength of the "set". This way one achieves an approximate logarithmic scaling of the wavelength points.

Once the "set" is established, the program identifies lines which contribute to the opacity in the set, both those that lie inside, as well as outside of the wavelength interval of the set. The parameters for these lines are stored in another arrays, which are obviously much smaller than the original ones.

The extent, and the wavelength points of the "set", are established by this procedure which involves the atomic line list. If molecules are included, the program goes through all the molecular line lists one by one, and analogously selects line that may contribute to the opacity.

The program then computes the opacity and emissivity at every wavelengths of the "set" and every depth, and if so desired, solves the radiative transfer equation to get the radiation intensity. Once this is done, the program establishes the next "set" with the starting wavelength being the last one of the previous set, and program repeats the procedure until the end of the requires spectral range, ALAST, is reached.

Such a straightforward scheme works well, but in order to minimize the computer memory requirements, SYNSPEC offers a procedure that is completely transparent to the user, and besides saving the computer memory it in fact also enables working with a large number of individual molecular line lists. It works as follows: If the number of selected lines from any line list is larger than MLINO (for atomic lines), or MLINMO (for molecular lines), the program does not stop, but instead continues an evaluation of the synthetic spectrum (or the opacity in the opacity table mode) to the wavelength at which the number of selected lines overpass MLINO or MLINMO. The program then reads the corresponding line list again, and selects new set of lines starting at this wavelength, minus the cutoff parameter CUTOFF. Since all the line lists are independent, it may happen that for some line lists this is never needed, but for some lists this has to be done repeatedly. For instance, for a cool star, the atomic line list and most of the molecular line lists produce the sets of selected lines that do not go over MLINO or MLINMO, but for instance for TiO the line list has to be read several times.

One can even set MLINO = MLINMO = 100,000, in which case many line lists are read several times. The computer time becomes somewhat larger, but a saving of computer memory is enormous, and even enables running SYNSPEC on some computers which may otherwise lack sufficient memory for certain applications. The procedure of re-setting the dimensions of corresponding large arrays associated with line lists will be described in detail in § 4.9.

3.6 Dealing with NLTE

There are several possibilities how to work with (at least approximate) NLTE:

• NLTE model atmosphere that was constructed by including the bound-free transition from a set of selected energy levels of selected ions and atoms, called *explicit levels*, free-free transition of explicit ions, and bound-bound transitions between explicit levels. The output model atmospheres, and thus an input to SYNSPEC, contains, besides temperature, electron density, and total density, also the NLTE level populations of all explicit levels. After a careful association of these levels with the levels referred to in the line list, these NLTE populations are being used by SYNSPEC to compute opacity and emissivity of lines of the explicit atoms and ions. Lines of species that were not explicit in the model atmosphere are treated by SYNSPEC in LTE. An example of this approach is using SYNSPEC to compute detailed spectra for model atmosphere of the OSTAR03 (Lanz & Hubeny 2003) and BSTAR07 (Lanz & Hubeny 2007) grids, as was done recently in Hubeny et al. (2025)

• Simplified NLTE model atmosphere that was constructed taking into account only a few species and energy levels as explicit. For instance, if an input model is nominally NLTE, and uses model atoms that are reasonably sufficient for some species (say H, Mg for a solar-type star), but uses very simple model atoms for metals, essentially with only very few low-lying levels, with the aim just to capture the most important contributions to the UV continuum opacity. While this approach would indeed reasonably describe the continuum opacity, it could lead to significant inaccuracies in the line opacity of these species.

To avoid these potential problems, SYNSPEC offers an option of **semi-explicit** species. These are those which are considered as *explicit* for evaluating the continuum opacity (that is, with their level population and bound-free cross-section evaluated using data from the TLUSTY atomic data files), while considered as *non-explicit* when evaluating the line opacity (that is, using data solely from the line list), and disregarding (potentially inaccurate) NLTE populations of their levels.

This option is invoked by setting the parameter MODE, which is a part of the second block of the standard input file, described in detail in Paper III, § 4.2, namely MODE=4. For convenience, we will provide a more detailed description of this parameter below.

• If an input model atmosphere was built using very detailed model atoms for, say, H, Mg, and Ca – these are true explicit species, and C, N, O, Na, and Si with simple model atoms – these may be treated by SYNSPEC as semi-explicit. No other species were explicit in the input model. However, we have realized, again as an example, that continua of Al, P, S, Fe, and Ni may provide an important contribution to the total continuum opacity. Since these species were not explicit in the input model atmosphere, their continuum opacity would not be considered by SYNSPEC at all. In order to remedy this problem, we have introduced the concept of **quasi-explicit species**. The user will provide atomic data files for them, which may be relatively simple, just to capture the bound-

free opacity from the most important levels. The populations of such levels will be computed in LTE, but their bound-free opacity *will* be included with the cross-sections determined by the input parameters from the corresponding atomic data files

• Finally, the input model atmosphere is an LTE model, but independent calculations provided NLTE departure coefficients (b-factors) for selected energy levels, and for basic atmospheric parameters equal or close to the parameters of input model atmosphere, Then one can take such LTE model as an input model atmosphere, and to use a file with the b-factors as an additional piece of information to obtain reasonably improved values of atomic level populations, and thus line profiles for these species. For instance, for a solar-type star, one may use an input LTE model atmosphere, and previously computed b-factors for, say, H, Ca, Mg (and perhaps others) to compute a spectrum that captures most of the observable NLTE effects.

4 Input data

4.1 Standard input file

As mentioned before, this is essentially a standard input file for TLUSTY. Although not all the parameters contained in the file are needed for SYNSPEC, using it is convenient because one does not need to construct a special standard input file for SYNSPEC. Moreover, it is usually supplied together with the model atmosphere file, `fort.8`. If the input model is not the one computed by TLUSTY, for instance that by Kurucz, or for opacity tables for "cool" temperatures (up to, say, 15,000 K), one may use the file `tab.5`, provided together with the program SYNSPEC that supplies necessary information. Briefly, the standard input file defines the chemical abundances, and supplies a list of "explicit" energy levels, i.e. those for which the bound-free transitions contribute to the opacity in the continuum. Notice that unlike spectral lines, where necessary information is contained in the line list(s), there is no separate "continuum" list that would contain necessary information about continuum processes. These have to be supplied in the standard input file, because the same information is needed for any model atmosphere construction.

For convenience, we display here the file `tab.5`. It is described in more detail in Papers I and III.

```
------------------------------------------------------------
6000. 4.0      ! TEFF, GRAV
 T  F              ! LTE,  LTGRAY
 ''
*
* frequencies
*
 2000
```

```
*
* data for atoms
*
 30             ! NATOMS
* mode    abn   modpf
    2      0.      0        ! H
    2      0.      0        ! He
    1      0.      0        ! Li
    1      0.      0        ! Be
    1      0.      0        ! B
    2      0.      0        ! C
    2      0.      0        ! N
    2      0.      0        ! O
    1      0.      0        ! F
    2      0.      0        ! Ne
    1      0.      0        ! Na
    2      0.      0        ! Mg
    2      0.      0        ! Al
    2      0.      0        ! Si
    1      0.      0        ! P
    2      0.      0        ! S
    1      0.      0        ! Cl
    1      0.      0        ! Ar
    1      0.      0        ! K
    2      0.      0        ! Ca
    1      0.      0        ! Sc
    1      0.      0        ! Ti
    1      0.      0        ! V
    1      0.      0        ! Cr
    1      0.      0        ! Mn
    2      0.      0        ! Fe
    1      0.      0        ! Co
    1      0.      0        ! Ni
    1      0.      0        ! Cu
    1      0.      0        ! Zn
*
* data for ions
*
*iat   iz   nlevs  ilast ilvlin  nonstd typion  filei
*
   1   -1     1      0      0      9    ' H- ' 'data/hmin.dat'
   1    0     9      0      0      0    ' H 1' 'data/h1.dat'
   1    1     1      1      0      0    ' H 2' ' '
   2    0    24      0      0      0    'He 1' 'data/he1.dat'
   2    1     1      1      0      0    'He 2' ' '
   6    0    40      0      0      0    ' C 1' 'data/c1.dat'
```

```
  6    1    22      0      0      0    ' C 2' 'data/c2.dat'
  6    2     1      1      0      0    ' C 3' ' '
  7    0    34      0      0      0    ' N 1' 'data/n1.dat'
  7    1    42      0      0      0    ' N 2' 'data/n2_32+10lev.dat'
  7    2     1      1      0      0    ' N 3' ' '
  8    0    33      0      0      0    ' O 1' 'data/o1_23+10lev.dat'
  8    1    48      0      0      0    ' O 2' 'data/o2_36+12lev.dat'
  8    2     1      1      0      0    ' O 3' ' '
 10    0    35      0      0      0    'Ne 1' 'data/ne1_23+12lev.dat'
 10    1    32      0      0      0    'Ne 2' 'data/ne2_23+9lev.dat'
 10    2     1      1      0      0    'Ne 3' ' '
 12    0    41      0      0      0    'Mg 1' 'data/mg1.t'
 12    1    25      0      0      0    'Mg 2' 'data/mg2.dat'
 12    2     1      1      0      0    'Mg 3' ' '
 13    0    10      0      0      0    'Al 1' 'data/al1.t'
 13    1    29      0      0      0    'Al 2' 'data/al2_20+9lev.dat'
 13    2     1      1      0      0    'Al 3' ' '
 14    0    45      0      0      0    'Si 1' 'data/si1.t'
 14    1    40      0      0      0    'Si 2' 'data/si2_36+4lev.dat'
 14    2     1      1      0      0    'Si 3' ' '
 16    0    41      0      0      0    ' S 1' 'data/s1.t'
 16    1    33      0      0      0    ' S 2' 'data/s2_23+10lev.dat'
 16    2     1      1      0      0    ' S 3' ' '
 20    0    15      0      0      0    'Ca 1' 'data/ca1.t'
 20    1    18      0      0      0    'Ca 2' 'data/ca2.t'
 20    2     1      1      0      0    'Ca 3' ' '
 26    0    30      0      0      0    'Fe 1' 'data/fe1.dat'
 26    1    36      0      0     -1    'Fe 2' 'data/fe2va.dat'
  0    0                                      'data/gf2601.gam'
                                              'data/gf2601.lin'
                                              'data/fe2p_14+11lev.rap'
 26    2    50      0      0     -1    'Fe 3' 'data/fe3va.dat'
  0    0                                      'data/gf2602.gam'
                                              'data/gf2602.lin'
                                              'data/fe3p_22+7lev.rap'
 26    3     1      1      0      0    'Fe 4' ' '
  0    0     0     -1      0      0    '    ' ' '
*
---------------------------------------------------------------
```

The file has a TLUSTY-compatible format. The lines beginning with "$*$" are understood as comments. The parameters `TEFF` and `GRAV` (the first line), which for TLUSTY have the meaning of the efefctive temperature $T_{\rm eff}$ and surface gravity $\log g$, do not have any meaning for SYNSPEC because the atmospheric structure, which those parameters would determine, is already given. Analogously, the parameter `NFREQ` (7th line) has no meaning for SYNSPEC.

The second block, "`data for atoms`", begins with a line with one or two numbers:

NATOMS - the highest atomic number of an element considered explicitly, i.e. the continua of which are being taken into account. We stress that, in contrast, lines of all element, regardless wether they are explicit or not, are being considered by SYNSPEC.

IABSET - adopted standard solar abundance set:

- $= 0$ – the adopted set of the standard solar abundances is that of Asplund et al. (2005), which is essentially the same as that of Grevesse et al. (2007) used in the recent large MARCS (Gustafsson et al. 2008) and Kurucz (Meszáros et al. 2012) model atmosphere grids.
- $= 1$ – the adopted set of the standard solar abundances is a combination of meteoritic/photospheric abundances from Asplund et al. (2009) adopted by Husser et al. (2013) in their extensive grid of PHOENIX model atmospheres and synthetic spectra.
- $= 2$ – the adopted set of the standard solar abundances is that of Grevesse & Sauval (1998), which was the default in the previous versions of TLUSTY and SYNSPEC.
- – any other number, or if IABSET is missing, IABSET is reset to 0.

Subsequently, there are NATOMS input records, each containing three numbers:

MODE – a specification of the mode of treatment for the element:

- $= 0$ – the element is not considered at all;
- $= 1$ – the element is treated implicitly, that is it does not contribute to the continuum opacity (but in the case of SYNSPEC it does contribute to the line opacity);
- $= 2$ – the element is considered fully explicitly; i.e. selected energy levels of the selected ionization stages are considered explicitly. In the case of SYNSPEC, these populations are either read from the input model atmosphere (in the case of NLTE models), or are computed by SYNSPEC (in the case of LTE models), using data provided by the atomic data files.
- $= 4$ – this is a new option, specific to SYNSPEC, setting the element to the *semi-explicit* mode, as described above.
- $= 5$ – this is another new option, specific to SYNSPEC, setting the element to the *quasi-explicit* mode, described above.

ABN – a specification of the chemical abundances of the individual species, in relation to the adopted standard solar abundance set, with the following convention:

- $= 0$ – the standard solar abundance is assumed
- < 0 – a non-solar abundance is assumed, abs(ABN) has now the meaning of the abundance expressed as a multiple of the standard solar abundance (i.e. -0.1 means $1/10$ of the standard solar abundance, -5 means 5 times standard solar abundance, etc.);

- > 0 – a non-solar abundance is assumed, expressed as $N(\text{elem})/N(\text{ref})$, i.e. relative by number to the reference species. The reference atom is H by default, but the reference species can be changed by means of the optional parameter IATREF (see § 7.4.4. of Paper III; and § 4.3.6 here).

MODPF – originally meant as a switch to set the mode of evaluation of the partition functions. However, the parameter MODPF is outdated because the evaluation of the partition functions (for atoms and ions) is now accomplished using Irwin (1981) data – see § 4.3.2.

The third bloc, "`data for ions`," specifies which atoms/ions are taken as explicit, which for SYNSPEC has the meaning that their bound-free and free-free opacities are taken into account: H^-, H I, He I, C I, C II, N I, N II, O I, O II, Ne I, Ne II, Mg I, Mg II, Al I, Al II, Si I, Si II, S I, S II, Ca I, Ca II, Fe I, Fe II, and Fe III. The highest ionization degree of each elements (e.g., H II, He III, etc., is considered as a one-level ion, just for a closure of the Saha-Boltzmann equations, so it does not provide any continuum opacity.

We do not describe here in detail the individual entries of this block; see Paper III, § 4.3. Very briefly: `iat` = atomic number; `iz` = charge, `nlevs` = number of explicit levels; `ilast` = switch to indicate whether the ion is the highest ion of the given species; `ilvlin, nonstd` – no meaning for SYNSPEC; `filei` = the file that contains detailed data for the given ion. All the files referred to in the file `tab.5` are a part of the standard distribution.

4.2 Input of keyword parameters

The current approach represents a formal change form the previous versions, but its introduction is motivated by an effort to simplify the input, and also make it more logical. We stress at the outset that the original scheme of input can still be used, so the users who already have available script files to work with the old-format input data may still use these scripts and files.

In the previous versions, there were two separate files designed to supply additional parameters that apply specifically for SYNSPEC. These files were the input file `fort.55`, and also the so-called "non-standard parameters" file, which, somewhat illogically, mixed the parameters used for generating the TLUSTY model, together with the parameters specific to SYNSPEC.

We now call all such parameters *keyword parameters*, or simply *keywords*. They all may be contained in the special input file called `kwords`. As mentioned above, if the user prefers, the keywords that were previously specified through the file `fort.55` may still be so specified, but for new applications it would be easier, and is recommended, to dispose of file `fort.55` altogether and use only file `kwords`.

An advantage of this system is that it is less prone to errors. The keywords are specified in the form `keyword name=value`, exactly as in the previous form of the non-standard parameters file. And, finally, a useful feature is that unlike

the input through file `fort.55`, not all parameters have to be explicitly specified because each is assigned a default value, appropriate for most applications.

It should be stressed that the previous non-standard parameters file name was specified in the standard input file (3rd line), typically as `'nst'` or `'params'`, or such. In the new system the 3rd line of the standard input file is still present (because this file is essentially a TLUSTY input file), but the file is disregarded, and SYNSPEC uses the keyword file that has to be named `kwords`.

4.2.1 Old system: Setting (some) keywords through `fort.55`

General structure of the file, using the variable names exactly as they are in the SYNSPEC source code, is as follows

```
IMODE   IDSTD   IPRIN
INMOD   INTRPL  ICHANG  ICHEMC
IOPHLI  NUNALP  NUNBET  NUNGAM  NUNBAL
IFREQ   INLTE   ICONTL  INLIST  IFHE2
IHYDPR  IHE1PR  IHE2PR
ALAM0   ALAST   CUTOF0  CUTOFS  RELOP   SPACE
```

up to here, these lines are mandatory; the rest is optional

```
NMLIST, (IUNITM(i),i=1,NMLIST)
VTB
NM0    ANG0    IFLUX
```

The parameters are read in a free format, and the last three lines of the file do not have to be present if they are not needed. In that case, `nmlist` and `nmu0` are set to 0, and `vtb` - the turbulent velocity - is specified in the non-standard parameters file anyway. These parameters are explained in detail in § 4.3.1.

4.2.2 New system: Setting all input keywords through the `'kwords'` file

The file is analogous to the so-called non-standard parameters file for TLUSTY. For instance, setting the parameters that were traditionally set through `fort.55` would look for instance like this:

```
IMODE=0,IDSTD=0,IPRIN=1
INMOD=1,INTRPL=0,ICHANG=0,ICHEMC=0
...
```
etc.

Each parameter is expressed by up to 6 characters; with letters being in upper case. The numerical values of the parameters must also have up to 6 characters. The separation of keywords is done through several possible special characters, but usually these are a carriage return, or a comma ",". As we specify below, all the parameters from the above example are actually set to the default values, so in fact nothing has to be specified in this case.

We also stress that if both files, `fort.55` and `kwords` are present, the values specified in `fort.55` take precedence over values specified in `kwords`[1].

4.3 Description of the keyword parameters

There are several types of keyword parameters, which will be described in turn.

4.3.1 Keywords optionally set by `fort.55`

First we describe the keywords that were previously set through `fort.55`. We stress again that these parameters can still be set up through the input file `fort.55`, but also through the keyword file `kwords`.

[**IMODE** – the basic mode of operation (for details, see § 4.1):
- = 0 - normal synthetic spectrum;
- = 1 - spectrum for a few lines (obsolete);
- = 2 - continuum (plus H and possibly He II lines) only;
- = −1 - an "iron curtain option" option;
- = −2 - "identification" option;
- = −3 - opacity table evaluation (with all lines);
- = −4 - opacity table with continua only.

DEFAULT; IMODE=0

IDSTD - standard depth:
- > 0 - index of the "standard depth", which is defined as a depth where $T \approx (2/3)T_{\rm eff}$. Approximately, `IDSTD` = $(2/3)ND$, with ND being the total number of depth points of the model. This parameter only influences the selection of lines to be considered.
- IDSTD= 0 – The set of "characteristic depth points" is composed of three depths, with indices 1, (ND-10)/2+1, and ND-9, i.e. id=1,ND,NDSTEP with NDSTEP=(ND-10)/2. This is the recommended option.
- IDSTD< 0 – The characteristic depths are set again as id=1,ND,NDSTEP, but now with NDSTEP=−IDSTD.Selecting IDSTD= −1 would lead to comparing opacities at all depth points. While this option would represent an overkill for classical atmospheres, it may be useful in special circumstances, for example for models with a chromospheric temperature rise, atmospheres with shocks, etc.

DEFAULT: IDSTD= 0

IPRIN – when > 0 increases the amount of printed information in the standard output.
DEFAULT: IPRIN=1

INMOD – an indicator of the input model:
- = 0 - the input model is a Kurucz model;

[1]This is done for a downward compatibility. If the user already has a working file `fort.55`, it may be kept, and only additional keywords would be specified through `kwords`.

• = 1 - the input is a TLUSTY model;
• = 2 - the input model is a model of a single accretion disk ring.
DEFAULT: INMOD=1
However, the new version of SYNSPEC recognizes whether the model is a TLUSTY or Kurucz model, so this parameters needs to be specified only if the input model is a TLUSTY model for an accretion disk ring.

INTRPL, ICHANG – now obsolete; if they are present in file `fort.55`, they have no effect.

ICHEMC – if non-zero, indicates a change of abundances with respect to the input model. In this case, an additional input file `fort.56` is required (see § 4.5).
DEFAULT: ICHEMC=0

IOPHLI - treatment of far Lα wings. Now obsolete, so the parameter has no effect.

NUNALP, NUNBET, NUNGAM, NUNBAL – if any of them set to a non-zero value, quasi-molecular satellites of Lα, Lβ, Lγ, and Hα, respectively, are taken into account, using old Allard & Koester data..
DEFAULTS: NUNALP=NUNBET=NUNGAM=NUNBAL=0

IFREQ – the choice of the radiative transfer equation solver; namely if `ifreq` $\geq$ 10 one uses the Feautrier scheme, otherwise the DFE scheme – see Paper II, § 3.8. This option is rarely used, except for pedagogical purposes.
DEFAULT: IFREQ=1

INLTE – if = 0 enforces LTE: lines are treated in LTE even if the input model is NLTE.
DEFAULT: INLTE=1

ICONTL – switch for including H and optionally He II lines (if IFHE2 $\neq$ 0) into the theoretical "continuum"
• = 0 – the lines are included in the continuum;
• = 1 – the lines are not include in the continuum.
DEFAULT: ICONTL=0

INLIST – specifies the format of the *atomic* line list:
• = 0 or 10, all the line lists are in text format,
• = 1 or 11, all the line lists are in binary format.
Notice that `inlist` has a dual meaning, because at the same time
• = 0 or 1 – the additional quantum numbers (see § 4.7.1), if present, are being taken into account;
• = 10 or 11, the additional quantum numbers, if present, are disregarded.
DEFAULT: INLIST=0

IFHE2 – if set to a non-zero value, He II is treated as a hydrogenic ion. Any data on He II lines from the line list are disregarded, and are replaced by

hardwired hydrogenic values.
DEFAULT: IFHE2=0

IHYDPR - mode of treating hydrogen Stark profiles:
- = 0 – approximate expression from Hubeny et al. (1994);
- = 1 – VCS profiles provided by Lemke (1997) tables;
- = 2 – using Tremblay tables from Tremblay & Bergeron (2009)
- < 0 – using Schoening & Butler (1989) tables

DEFAULT: IHYDPR=0

IHE1PR – mode of treating He I lines:
- = 0 - broadening parameters taken from the line list;
- > 0 - broadening parameters for 4 triplet lines,$\lambda\lambda$ 4026, 4387, 4471, and 4921, are treated using special line broadening tables, after Barnard et al. (1974) and Shamey (1969).

DEFAULT: IHE1PR=0

IHE2PR – mode of treating He II line profiles:
- = 0 – default way, i.e.

- if IFHE2 > 0, using the approximate hydrogenic Stark + Doppler profile after Hubeny et al. (1994);
- if IFHE2 = 0, they are treated as any other metal lines with the broadening parameters specified in the line list.

- > 0 the He II line profiles are given through the Stark broadening tables of Schoening & Butler (1989)

DEFAULT: IHE2PR=0

ALAM0 – starting wavelength [Å]. This parameter has no natural default value, and should be always supplied by the user. Setting it to zero means that the synthetic spectrum is computed from the wavelength of the first line present in the atomic line list.
DEFAULT: ALAM0=0.

ALAST – its absolute value is the ending wavelength [in Å] of the synthetic spectrum,
- if ALAST< 0, then all the line wavelengths are in vacuum (otherwise by default all wavelengths above 2000 Å are taken as the air wavelengths). Again, there is no natural default value; it should be set according to the chosen application.

DEFAULT: ALAST=10000.

CUTOF0 – – cutoff parameter: any line except H and He II is allowed to contribute up to a distance CUTOF0 Å from its center.
DEFAULT: CUTOF0=20.

CUTOFS≡NFREQC – the parameter was originally called CUTOFS, but was kept only for keeping the original format of file `fort.55` unchanged. Its original meaning was a cutoff for auto-ionizing lines. It has now the

meaning of number of frequency points in the continuum. The opacity in the continuum in these points is used only for rejecting weak lines (together with the parameter RELOP - see below). The continuum frequencies are spaced logarithmically equidistantly between ALAM0 and ALAST. It should be stressed that these continuum frequencies are used **only** for the purpose of line rejection[2].
DEFAULT: NFREQC=2000

RELOP – line rejection parameter: a line is rejected if the ratio of its line-center opacity to the continuum opacity at all standard depths (set through IDSTD) is less than RELOP.
DEFAULT: RELOP=1.e-4

SPACE – spacing parameter, which represents the maximum distance of two wavelength points at the midpoint of the considered wavelength interval; the actual maximum spacing is proportional to the wavelength. For more details, refer to § 3.5, or Paper I.
DEFAULT: SPACE=0.01

NMLIST – – number of molecular line lists. But this can be overwritten by the input from file `fort.3`, if present.
DEFAULT: NMLIST=0

IUNIM1, IUNIM2 – unit numbers for the molecular line lists; applies only if the lists, (no more than 2 of them), are specified through these keywords, and not through an input file `fort.3`, which is much more flexible, and thus recommended.
DEFAULTS: IUNIM1=20, IUNIM2=21

VTB – turbulent velocity [km s^{-1}].
DEFAULT: VTB=2.

NMU – if set to a non-zero value, SYNSPEC produces not only the flux, but also the specific intensities, $I_\nu(\mu)$ [erg $cm^{-2}s^{-1}Hz^{-1}ster^{-1}$]; μ is the cosine of the angle with respect to the normal to the surface. The value of NMU represents the number of the angle points, expressed as μ, and set equidistantly between ANG0 and 1.
DEFAULT: NMU=0

ANG0 – minimum angle cosine. It has a meaning only if NMU > 1.
DEFAULT: ANG0=0.1

IFLUX – should be set to 1 to invoke an evaluation of specific intensities;
DEFAULT: IFLUX=0

[2] Actual continuum opacities and radiation fluxed are computed on much finer frequency mesh, namely as edge frequencies of each "set", i.e their spacing is about 1 Å.

To avoid confusion, we stress that in the original `fort.55` file, the unit numbers for the molecular line lists are being read as

```
read(unit,(amlist(i),i=1,nmlist))
```

which formally allows for an arbitrary number of lists. However, the `kwords` file allows only for up to 2 molecular line lists. If more lists are needed, it is strongly recommended to use additional input file `fort.3` to specify the molecular line lists, in which case the information from `fort.55` or `kwords` is disregarded.

In the following, we will describe keyword parameters that can be set up, if needed, only from the file `kwords`.

4.3.2 Keywords for setting the treatment of molecules

IFMOL – basic switch to include molecules:

- $= 0$ – molecules are disregarded;
- > 0 molecules are taken into account.

DEFAULT: IFMOL=1 if the effective temperature $T_{\rm eff} \leq 9000$ K; otherwise IFMOL=0.

TMOLIM – limiting value of local temperature to consider molecules; i.e. molecules are disregarded for $T >$ TMOLIM even if IFMOL is set to 1.
DEFAULT: TMOLIM=9000.

IFEOS – a switch to activate the "EOS-only" option.

- $= 0$ – the standard option to compute either a synthetic spectrum or an opacity table, depending on the input parameters;
- > 0 – an "EOS-only" option, in which SYNSPEC works as in the mode of evaluating the opacity table for a set of temperature/density pairs, but instead of computing the opacities, it just solves for the global chemical equilibrium (i.e., the equation of state - EOS) to obtain number densities of all atomic and molecular species.
- < 0 – the EOS quantities and output files are produced for an otherwise standard run (synthetic spectrum or opacity tables), and the quantities are printed for a loop in depth indices ID=1,ND, abs(IFEOS). For the opacity tables, there is formally only one depth, so in order to print the EOS quantities one sets IFEOS= -1. For synthetic spectra, setting for instance IFEOS=-30 activates printing the EOS quantities at depths 1, 31, 61, etc.

DEFAULT: IFEOS=0

MOLTAB - a switch to select the Tsuji-like table for solving for the chemical equilibrium (EOS).

- $= 0$ – original Tsuji (1973) table;
- $= 1$ – modified table with added data from Barklem & Collet (2016), transformed by setting polynomial fits analogous to the original Tsuji fitting formula. This option is strongly recommended.

DEFAULT: MOLTAB=1

IRWTAB - a switch to select the table of fitting coefficients to evaluate the molecular partition functions;
• = 0 – the original Irwin's table is used. Since the improved table is obviously preferable, we keep this option for possible comparisons with old versions.
• ≠ 0 – an upgraded table, supplemented by Barklem & Collet (2017) data for diatomic molecules, transformed to polynomial expansion analogous to that of Irwin, is used.
DEFAULT: IRWTAB=1

IPFBC – if ≠ 0, detailed BC tables for partition functions are used instead of polynomial fits of the upgraded Irwin-like table. If this keyword is set, it takes precedence over the specifications through IRWTAB, for molecules for which the BC tables are available.
DEFAULT: IPFBC=1

IPFEXO - a switch to include the EXOMOL partition functions for the available molecules:
• = 0 – the EXOMOL partition functions are not considered;
• ≠ 0 – the EXOMOL partition functions are considered, and override other values whenever available.
If the keyword is set, it takes precedence over all previous specification, for molecules for which the EXOMOL tables are available.
DEFAULT: IPFEXO=1

IEQBC – if ≠ 0, detailed BC tables for equilibrium constants are used instead of polynomial fits of the upgraded Tsuji-like table, for molecules for which the BC tables are available.
DEFAULT: IEQBC=1

GSSTD – the parameter of the "standard" value of the Stark broadening coefficient for molecular lines.
DEFAULT: GSSTD=1.e-5

GWSTD – the parameter of the "standard" value of the Van der Waals broadening coefficient for molecular lines, see Eq.(29) of Paper V
DEFAULT: GWSTD=1.e-7

4.3.3 Association of energy levels with the input NLTE model atmosphere

Next group are the keywords influencing the use the additional quantum numbers contained in special atomic line lists, which are meant to improve an association between the atomic energy levels given in the input model atmosphere, and those referred to in the line list. For details, see Paper IV, § 3.6. This approach is used for light elements. For iron-peak elements, one uses a different strategy. This is desrcibed in § 4.4.

The following keywords control the use of the energy and quantum number limits which are the part of some standard atomic data files.

ERANGE – resets the energy limits ENION1 and ENION2, that are specified in the atomic data file, in the case when they are set too close to the actual energy of the level ENION. Specifically, if (ENION1-ENION)/ENION$< 10^{-6}$ then ENION1 is reset to ENION1=ENION (1-ERANGE); and analogously ENION2=ENION (1+ERANGE). DEFAULT: ERANGE=0.1 (i.e., 10%)

ISPICK – if set to 0, then the S quantum number is not used for associating the levels from the line list to the explicit levels
DEFAULT: ISPICK=1 (i.e., the S value is being used)

ILPICK – analogous switch for the L quantum number
DEFAULT: ILPICK=1 (i.e., the L value is being used)

IPPICK – analogous switch for the parity
DEFAULT: IPPICK=1 (i.e., the parity is being used).

4.3.4 Hydrogen line profiles

Next group are the keywords for additional options in treating hydrogen line profiles. Previously, the main keyword was IHYDPR, described in § 4.2.1 The original treatment also allowed for including the opacity of the quasi-molecular satellites of Lyman α, β, γ and Balmer α through keyword parameters NUNALP, NUNBET, NUNGAM, NUNBAL, see § 4.2.1, using older data from Allard & Koester (1992).

The newly introduced keyword parameters are the following:

IHYDDK – a switch to include newer D. Koester's data taken from his code ATMDK.
- $= 0$ – the Koester's newer data are not used. The treatment of hydrogen lines remains being controlled by other keywords.
- $\neq 0$ – line broadening of Lα and Lβ is treated using Koester's tables that include broadening of these lines by neutral hydrogen, protons, H_2, and electrons. In this option, the previous setting switched on by keywords IHYDPR, as well as NUNALP and NUNBET are disregarded.

DEFAULT: IHDDK=0 (it should be set to 1 for white dwarfs, but for other stars it would have a negligible effect, and would unnecessarily complicate an evaluation of the opacity.

IHGOM - switch for treating line broadening of Lα, β, γ using tables from Gomez et al.(2021) using the code Xenomorph.
- $= 0$ – the Xenomorph tables are not used;
- $\neq 0$ – the Xenomorph tables are used to compute the total cross-section for absorption by the ground-state neutral hydrogen between 942 and 1478 Å. The table combines absorption in the Lyman lines together with the

pseudo-continuum opacity in between. This option takes precedence over any other option switched on by `NUNALP, NUNBET, NUNGAM`, and IHDDK.
DEFAULT: IHGOM=0

HGLIM – lower limit of electron density for which the Xenomorph data are taken into account (that is, if `IHGOM` is set to 1).
DEFAULT: HGLIM=1.e18 (so the data are used only for $n_{\rm e} \geq 10^{18}{\rm cm}^{-3}$).

There is an additional keyword parameter for treating the broadening of Lα by neutral helium; better speaking, the H-He collision-induced satellites in the in the Lα profile from Allard et al. (2020) (important for DBA white dwarfs)

NUNHE – if $\neq 0$, the H-He collision-induced satellites in the in the Lα profile is included.
DEFAULT: NUNHHE=0

4.3.5 Additional opacities

As in the previous versions, there is a number of keywords that switch on an evaluation of particular additional opacities. They are summarized in Table 1.

IOPHMI, IOPH2P, IOPHEM, IOPH2M, IOPCH, IOPOH – switches to include additional continuum opacities.
DEFAULT: all are = 1

IOH2H2, IOH2HE, IOH2H1, IOHHE - switches to include CIA (collision induced absorption) opacities.
DEFAULT: all are = 1 if IFMOL=1 (i.e. if molecules are included).

IRCST, IRSCH2, IRSCHE – switches to include Rayleigh scattering on H I, H_2, and He I, respectively.
DEFAULT: all are =1 for IMODE ≥ -2 (spectrum synthesis),
and are = 0 for IMODE ≤ -3 (opacity table evaluation)

4.3.6 Additional keywords

Finally, there are several additional keywords:

IATREF – index of the "reference" atom, which is the one to which the chemical abundances are related to. This keyword has only a very limited importance for SYNSPEC. For instance, for a DBA star with very small traces of hydrogen, He can be chosen a a reference atom, in which case IATREF=2
DEFAULT: IATREF=1, i.e. the index of the first explicit atom, usually hydrogen.

NFREQS – number of frequency points in a "set", an elementary frequency range for solving the transfer equation – see § 3.5 for details. Again, its value has now a limited importance, and the default value is essentially

Table 1: Additional opacities in SYNSPEC – keywords and their default values

Opacity source	Keyword	Default	Set automatically to 0 if
H- b-f and f-f	IOPHMI	1	H- is already an explicit ion
H_2^+ b-f and f-f	IOPH2P	1	IFMOL=0 or $T >$ TMOLIM
H_2^- f-f	IOPH2M	1	IFMOL=0 or $T >$ TMOLIM
He- b-f and f-f	IOPHEM	1	
CIA H_2-H_2	IOH2H2	1	IFMOL=0 or $T >$ TMOLIM
CIA H_2-He	IOH2HE	1	IFMOL=0 or $T >$ TMOLIM
CIA H_2-H	IOH2H	1	IFMOL=0 or $T >$ TMOLIM
CH opacity	IOPCH	1	IFMOL=0 or $T >$ TMOLIM
OH opacity	IOPOH	1	IFMOL=0 or $T >$ TMOLIM
H Rayleigh scat.	IRSCT	1	IMODE= -3 (opacity table)
He Rayleigh scat.	IRSCHE	1	IMODE= -3 (opacity table)
H_2 Rayleigh scat.	IRSCH2	1	IFMOL=0 or $T >$ TMOLIM or IMODE= -3 (opacity table)

always satisfactory.
DEFAUL:T: NFREQS=120

IBFAC – a switch defining the meaning of the input NLTE populations in the input file `fort.8`:
• = 0 – the input file contains the absolute level populations (in addition to the temperature, electron density, mass density, and, for models with molecules, the total particle number density) – the standard option;
• = 1 – the input file `fort.8` contains the b^*-factors instead of absolute NLTE populations. When using this option, the user has to make sure that the file `fort.8` indeed contains the absolute b^*-factors, not the usual b-factors[3].
• = 2 – In this case the file `fort.8` is an input LTE model atmosphere (just T, n_e, ρ), and the b^*-factors are supplied from an additional input file named `bfactors`. For details, refer to Appendix B.
DEFAULT: IBFAC=0 (i.e., the standard option)

IDIFF - a switch to compute depth-dependent radiative acceleration of all atoms in all ionization stages up to the 6th
• = 1 - the radiative accelerations are computed;
• ≠ 1 - the radiative accelerations are not computed
DEFAULT: IDIFF=0

[3]We distinguish two types of b-factors since both are used in the literature. The usual b-factor is the ratio of the actual population of the LTE population defined with respect to the ground state of the next ion, while the absolute b^*-factor is defined as a ratio to the LTE population taking into account all ionization stages. For details refer to Appendix B

4.4 Additional file `superlevels`

In NLTE metal line-blanketed models, such as those in the grids OSTAR03 and BSTAR07 (Lanz & Hubeny 2003, 2007), one does not deal with individual energy levels of iron-peak elements, such as Fe or Ni, but by means of the so-called superlevels. As described in detail in Hubeny & Lanz (1995), a superlevel is a set of energy levels with the same parity, and with relatively close level energies.However, a span of energies of levels forming a superlevel may still be relatively large. Therefore, one needs to supply an additional input file called `superlevels` that contains an information about the energy limits of superlevels of both parities. The structure of such file is explained on the following example, which provides a necessary input file when producing synthetic spectra for the BSTAR2007 model atmospheres. We display here only the beginning the file; the complete file is a part of the standard distribution.

```
-----------------------------------------------------------------
  4            ! number of ions with superlevels
 26 1          ! atomic number and charge of the 1st ion  (Fe II)
23             ! number of even-parity superlevels
  1000.        ! upper energy limit for the 1st superlevel
  5000.        ! analogously for other superlevels
 10000.
 20000.
 25000.
 30000.
 35000.
 40000.
 45000.
 52000.
 65000.
 75000.
 82000.
 90000.
101000.
112000.
131000.
150000.
165000.
180000.
200000.
220000.
250000.
13            ! number of odd-parity superlevels
 40000.       ! energy limits, analogously as before
 50000.
 55000.
 70000.
```

```
 80000.
100000.
116000.
131000.
155000.
180000.
210000.
240000.
280000.
------------------------------------------------------------------
```

This section contains data for Fe II. The complete file `superlevels` contains analogous dat for Fe III, IV, and V.

4.5 Changing chemical abundances – File `fort.56`

File `fort.56`, which is needed only if the keyword `ICHEMC` is set to a non-zero value, has a simple structure. The 1st line contains

```
nchang
```

and then `nchang` records, each with

```
iatom  abn
```

where

• `nchang` – number of chemical elements for which the chemical abundances are changed with respect to the TLUSTY input (specified in the standard input file);

• `iatom` – atomic number;

• `abn` – the modified abundance. The convention is as follows:

`abn` > 0 – abundance by number, relative to hydrogen; if ABUN > 1, the abundance expressed as $12 - \log(N/N_H)$, as is often used in observational studies.

`abn` $= 0$ – the abundance is solar;

`abn` < 0 – the absolute value is the ratio of the abundance to the one specified in the standard input.

For all so specified chemical elements, the values of `abn` represent the current chemical abundances for the run of SYNSPEC. For the explicit, NLTE, species, they are also used to multiply the NLTE level populations given by the models atmosphere input by the ratio of the "new" abundance of the corresponding chemical element (specified by the values in the file `fort.56`), and the "old" abundance, referred to in the standard input file, that were used previously by TLUSTY. This procedure is not rigorous because the NLTE line formation is not a linear process, and the modified abundances may yield a different global atmospheric structure. But this is acceptable if the relative difference of the old and new abundances is small, say, no more than 0.2 to 0.3 dex. For larger abundance differences, it is more prudent first to recompute the model atmosphere and then to produce a synthetic spectrum.

4.6 Setting the names and formats of line lists - file `fort.3`

In the previous versions, there were two options to specify line lists: (i) to specify unit numbers in the file `fort.55`, so that the line lists need to be copied or linked to files named `fort.xx` (by default, xx=19 for an atomic line list, and the defaults for two molecular lists are 20, 21); or (ii) to set up file `fort.3` which refers to the filenames of the line lists directly. This is described below.

In this version, both options are still available, but it is strongly recommended to specify the line lists to be used through file `fort.3`, which is more flexible. Using just `fort.55` (or, in fact, `kwords`) is useful only if one or two molecular line lists are considered, while using file `fort.3` allows the user to have many line lists, typically a separate list for each individual molecule. Using this option, one can generate the individual line lists (as shown in Appendix C), and use them without merging them into one big list which is more cumbersome to handle. In this option, one can also easily explore a relative importance of the individual molecular species in the resulting synthetic spectrum.

Another modification is that SYNSPEC can now correctly deal with molecular ions, which was not the case in the previous versions.

An example of the `fort.3` file is the following one[4]

```
'gfATO.dat'       0
'mol/MgH.list'    0   9000
'mol/MgO.list'    0   9000
'mol/CH.list'     0   9000
'mol/CH+.list'    0   9000
'mol/C2.list'     0   9000
'mol/CN.list'     0   9000
'mol/CO.list'     0   9000
'mol/AlO.list'    0   9000
'mol/AlH.list'    0   9000
'mol/CaO.list'    0   9000
'mol/CaH.list'    0   9000
'mol/OH.list'     0   9000
'mol/OH+.list'    0   9000
'mol/HF.list'     0   9000
'mol/SiO.list'    0   9000
'mol/SiH.list'    0   9000
'mol/FeH.list'    0   9000
'mol/NS.list'     0   9000
'mol/NO.list'     0   9000
'gfTiO.bin'       1   5000
```

The files `gfATO.dat` (atomic line list), and `gfTiO.bin` (TiO line list in the binary format) were already a part of the previous distribution presented in

[4]The partial line lists referred to in this example actually exist, and may be provided upon request. But they can also be easily constructed by the user using the programs described in Appendix C.

Paper IV. The above example assumes that there is a subdirectory `./mol` in which the individual molecular line list are stored. The second entry on each line indicates whether the list in the text (ASCII) format (0), or the binary format (1). The last entry specifies the maximum temperature for which the line list is considered. However, this takes effect only for an opacity table construction; for a regular spectrum synthesis all the lists are considered anyway, and which lines are selected to be included in the spectrum synthesis is determined by the procedure described in Paper V, and here in Appendix A.2.

Besides the possibility of directly distributing line lists to interested users, we now offer another option, namely a suite of short programs which allow the user to generated SYNSPEC compatible lists directly from the EXOMOL database. This also offers a flexibility of generating lists of subsets of stronger lines, or lists in a particular wavelength range. These programs are described in Appendix C.

4.7 Structure of the line lists

Line lists are at the very heart of the spectrum synthesis provided by SYNSPEC. Several variants are available in the standard distribution of SYNSPEC. A complication here is that the lists may contain data for many millions of lines (atomic lists), and literally billions of lines for molecules. Therefore, some choices have to be made by the user, in view of required accuracy of the resulting spectra, and also with the consideration of the memory and time constraints on available computer resources.

4.7.1 Atomic line lists

The line lists were described in Paper V, but for convenience we present a brief synopsis here.

Currently, there are two variants of an atomic line list, the *standard* one, used in the previous versions of SYNSPEC (before 1.54), and an *extended* one, which is preferable for NLTE model atmospheres because it provides a better association between atomic energy levels used in TLUSTY and SYNSPEC.

(*i*) The *standard line list* contains one input record for each spectral line. In the past there could be an additional record, but this option is cumbersome and was rarely used, so it is not supported any longer. Each input record contains the following variables:

```
alam anum gf excl ql excu qu agam gs gw inext
```

These quantities have the following meaning:

• `alam` – wavelength [nm]. One follows the convention that for $\lambda < 200$ nm the wavelengths are taken in vacuum, while for $\lambda \geq 200$ nm the wavelengths are for standard air. These wavelengths can however be transformed to the vacuum wavelengths when computing actual spectra by setting the keyword `ALAST` (see § 4.3) to a negative value;

• `anum` – numerical code of the element and the ionization degree (using Kurucz's

convention, namely atomic number, decimal point, and degree of ionization: for instance 2.00 = He I, 6.03 = C IV, 26.01 = Fe II, etc.)
• `gf` – $\log gf$ (g is the statistical weight of the lower level, and f the oscillator strength)
• `exclu` – excitation potential of the lower level [cm^{-1}]
• `ql` – the J quantum number of the lower level
• `excu, qu` – analogous quantities for the upper level
• `agam` – if non-zero, $\log \Gamma_{\mathrm{rad}}$ for radiation damping [s^{-1}]
• `gs` – if non-zero, $\log \Gamma_{\mathrm{stark}}$ for Stark broadening [s^{-1}]
• `gw` – if non-zero, $\log \Gamma_{\mathrm{vdW}}$ for Van der Waals broadening [s^{-1}]
• `inext` – it is now a dummy parameter, since its previous meaning, namely invoking an addition record for a current spectral line, is not supported any longer.

If any of the parameters `agam, gs, gw` is set to zero, approximate formulae are adopted for evaluating the corresponding broadening parameter. For details refer to the Paper V, § 5.1.

(*ii*) The *extended line list*. The entry for each spectral line contains, in addition to the standard 11 parameters the set of 6 following new parameters:

```
isql ilql ipql isqu ilqu ipqu
```

where
• `isql` – the quantum number $2S + 1$ of the lower level
• `ilql` – the quantum number L of the lower level
• `ipql` – the parity of the lower level
• `isqu, ilqu, ipqu` – analogous values for the upper level.
These parameters are explained in more detail in Paper IV § 3.6. We stress that the atomic line list provided in the current distribution, file `gfATO.dat`, or its binary version `gfATO.bin` are *extended* lists.

4.7.2 Molecular line lists

As described in the Paper V, §5̇.2, there are three variants of the molecular line list; the *standard* (classical-broadening) list; its variant called a *minimalistic* list; and the EXOMOL-*type* list, which contains additional Van der Waals broadening parameters, described in detail in Paper V, § 5.2.
We describe the individual types of lists below.

(*i*) *Standard list*. Its structure is analogous, albeit simplified, to a standard atomic list. Each input record contains the following parameters for each line:

```
alam anum gf excl agam gs gw
```

The meaning of the variables is exactly equivalent to those for the atomic line list. Since the molecular lines one always assumed to be formed in LTE, the parameters for the upper level (EXCU and QU) are not needed. The largest

source of uncertainty in treating lines whose data are provided by such a list is in the evaluation of pressure (Van der Waals) broadening. We stress that Stark broadening is typically negligible under the conditions where molecules are formed in significant numbers to make molecular lines contribute appreciably to the total opacity, and natural broadening is usually well approximated by the classical formula. The Van der Walls broadening parameter is expressed by Eq. (29) of Paper V.

(*ii*) A *minimalistic list*. Its introduction was motivated by the observation that in a vast majority of molecular line lists the parameters `gs` and `gw` have the same value for all lines, and moreover for most molecular species. Therefore, for such molecules it is a waste of computer memory to store them in the line list; instead they are assigned by SYNSPEC when initializing line data from the line list. Similarly, the natural broadening in such lists has its classical value, which again does not have be stored in the line list, but is computed by SYNSPEC instead. Consequently, the minimalistic line lists contain only four values for each line:

```
alam anum gf excl
```

which have the same meaning as above. The parameter `agam` is computed as `agam` $= 2.2e13/\lambda^2$, with the wavelength λ expressed in nm. Parameters `gs` and `gw` can be specified throughthe keyword parameters `GSSTD` and `GWSTD`, which have default values 3.1e-5 and 1.0e-7, respectively – see § 4.3.2. Therefore, if they are not specified in the input, the parameters `gs` and `gw` will have the same values as in the usual standard lists.

(*iii*) The EXOMOL-type list contains for each line the following parameters:

```
alam anum gf excl agam  gvdwh2 gexph2 gvdwhe gexphe
```

where the last 4 parameters describe the Van der Waals broadening following Barton et al. (2017a,b) as specified by Eq. (30) of Paper V.

We stress that SYNSPEC recognizes which type of the list is used, so that no additional input parameter needs to be specified.

In the previous release of SYNSPEC, described in Paper IV, there were 3 molecular line lists available: `gfMOL.dat`, `gfTiO.dat`, and `gfH2O.dat`, together with their binary versions `*.bin`. The first one, `gfMOL`, contains data for the following 18 molecules: H_2, CH, NH, OH, NaH, MgH, SiH, CaH, CrH, FeH, C_2, CN, CO, MgO, AlO, SiO, CaO, and VO, taken mostly from the Kurucz website [5]. The other two line lists are for TiO and H_2O, respectively, as their names imply.

The recent development is that there may be many more molecular line lists for individual molecules, or a small groups of molecules. The plan is to set up line lists for all molecules, and molecular ions, for which the EXOMOL data are available. Since it may be cumbersome, and memory intensive to made them

[5] http://kurucz.harvard.edu/molecules

all available for downloading, we present instead a suite of simple programs designed to extract data directly from the EXOMOL database, and to set up individual line lists in the SYNSPEC compatible format. This is described in Appendix C.

It should be noted that EXOMOL considers the individual isotopologues of a molecular species separately, while SYNSPEC considers a given molecule as a combination of all its isotopologues with the solar isotopic ratios. Similarly, the partition function of a molecule is given as an appropriate average over the partition function of all its isotopologues. This is accomplished by short programs PFSUM and LIST, described in Appendix C. We also stress that the line-list generating program LIST produces a *minimalistic* list since the detailed broadening parameters are not given for most of the molecular species, or for stellar conditions. For the species for which EXOMOL provides data for line broadening, a more sophisticated approach has to be adopted (as was done earlier for H_2O), but this will depend on an actual molecule. Such an endeavor will be considered in a future paper.

4.8 Opacity tables - file `fort.2`

As specified above, the option of computing opacity table is switched on by the basic keyword `IMODE`. If it is set to -3 or -4, an additional input file `fort.2` is required. Its composition has been described in Paper IV, but for convenience we repeat it here.

The input file `fort.2` has at least 5 lines, but in some cases (see below) more:

- **1st line**: NTEMP, TEMP1, TEMP2:

NTEMP – number of temperatures, and the mode of setting the opacity table:
- > 0 – the opacity table is computed in the classical mode; NTEMP represents the number of temperatures;
- $= 0$ – the opacity table is computed in the model-tailored mode. In this case the temperatures, densities, and electron densities are read from input model, stored in the file `fort.8` as in the standard run of SYNSPEC. The remaining parameters on this line and the two next lines do not take effect.

TEMP1 – lowest temperature [K];

TEMP2 – highest temperature [K].

In the classical mode, the temperatures are set to be logarithmically equidistant.

- **2nd line**: IDENS:

IDENS – A switch that specifies the treatment of density, and the nature of the density parameter. It has a dual meaning:
(i) the value of mod(IDENS, 10) determines the nature of the density parameter:

• mod(IDENS, 10)= 0 – (i.e. IDENS = 0, 10, 20) – the density parameter has the meaning of electron density;
• mod(IDENS, 10)= 1 – (i.e. IDENS = 1, 11, 21) – the density parameter has the meaning of mass density;
• mod(IDENS, 10)= 2 – (i.e. IDENS = 2, 12, 22) – the density parameter has the meaning of gas pressure.

At the same time, the value of IDENS determines
(ii) the actual values of the density parameter to be used in the opacity table. The next block of input depends on this value:

- IDENS < 10 – (i.e. IDENS = 0, 1, 2) – the values of the density parameters are the same for all temperatures. In other words, the pairs of temperature/density form a regular rectangle in the $\log T/\log \rho$ space (for simplicity, we take here the mass density for the density parameter, but the situation is the same for all possible choices of the density parameter). In this case the next block contains just one record:

  — **3rd block – one line**: NDENS, DENS1, DENS2, where:

  NDENS – number of densities;

  DENS1 – lowest density parameter [cgs units];

  DENS2 – highest density parameter [cgs units].

- IDENS ≥ 10 and IDENS < 20 – similar to above, but the T/ρ pairs occupy a slanted rectangle in the $\log T/\log \rho$ space. One specifies the limiting values separately for the lowest and for the highest temperature. For other temperatures, the limiting vales are obtained by a logarithmic interpolation. In this case, the next block also contains one record, namely:

  —**3rd block – one line**:
  NDENS, DENSL1, DENSL2, DENSU1, DENSU2, where:

  NDENS – number of densities;

  DENSL1 – low density parameter limit for the lowest temperature [cgs units];

  DENSL2 – high density parameter limit or the lowest temperature [cgs units].

  DENSU1 – low density parameter limit for the highest temperature [cgs units];

  DENSU2 – high density parameter limit or the highest temperature [cgs units].

- IDENS ≥ 20 and IDENS < 30 – the density limits and the number of densities are generally different for each temperature. In other words, the T/ρ pairs occupy a completely irregular area in the $\log T/\log \rho$ space. In this case, the next block contains NTEMP input records, each containing the limiting values as before:

— **3rd block – NTEMP lines**: for each T there are three values,

NDENS, DENS1, DENS2, where:

NDENS – number of densities for the given temperature;
DENS1 – lowest density parameter [cgs units] for this T;
DENS2 – highest density parameter [cgs units] for this T.

Again, the densities for each temperature are spaced logarithmically equidistantly.

We stress that for cool stars, it is most advantageous to use mass density (e.g., IDENS=1, 11, or 21) as the density parameter.

- **4th block – one line**: NLAMB, INTTAB, WLAM1, WLAM2:

NLAMB – number of wavelength points in the final (universal) opacity table;

INTTAB - a switch for determining the mode of transformation of opacities from the internal to the universal wavelength grid points:
$= 1$ – the opacities in the universal grid are determined by an interpolation in wavelength;
$\neq 1$ – the opacities in the universal grid are determined by approximately preserving the integral over wavelength - see Eq. (1) below.

WLAM1 – shortest wavelength in the opacity table [Å];

WLAM2 – longest wavelength in the opacity table [Å].

- **5th block – one line**: TABNAM, IBINGR:

TABNAM – a character variable specifying the name of the file where the resulting opacity table is stored;

IBINGR – the format of the opacity table:
- $= 0$ – the table is stored in text format;
- > 0 – the table is stored in binary format.

For example, a simple file `fort.2` that can be used as a test, is

```
------------------------------------------------------------------
11  3.0e3    1.5e4
1
7  1.0e-12  1.0e-6
100000  0  900. 110000.
'optab11_7f.dat'    0
------------------------------------------------------------------
```

which constructs a table for 11 temperatures between 3000 and 15000 K, 7 values of mass density between 10^{-12} and 10^{-6} g cm^{-3} (both logarithmically spaced, and forming a regular grid), for 100,000 wavelengths logarithmically spaced between 900 and 110,000 Å. The table is stored in the file `optab11_7f.dat`, in the text format.

If one intends to construct a table (as an example) with the same density limits as before for the lowest T (3000 K), but with limits 10^{-10} and 10^{-4} at the highest T, and with interpolated values of the density limits in between, the first three lines of `fort.2` will look like:

```
------------------------------------------------------------------
11  3.0e3    1.5e4
11
7  1.0e-12  1.0e-6  1.e-10   1.e-4
 ------------------------------------------------------------------
```

When the interpolation option `INTTAB` is set to 1, the tabular opacity is evaluated as

$$\kappa^{\rm tab}(\lambda_i) = \frac{2}{\lambda_{i+1} - \lambda_{i-1}} \int_{(\lambda_i+\lambda_{i-1})/2}^{(\lambda_i+\lambda_{i+1})/2} \kappa(\lambda)\, d\lambda, \qquad (1)$$

where $\kappa(\lambda)$ is the (detailed) opacity computed for the internal wavelength points.

The next portion of the file `fort.2` is optional. If it exists, it contains information about changing chemical abundances or removing selected species. It should be stressed that such changes concern only the evaluation of opacities, and are meant as a tool for producing an opacity table for use in the hybrid NLTE/opacity-table mode of TLUSTY, where some species are treated explicitly, while the opacity of the remaining species enters through a specific opacity table that does not contain opacity from the species to be treated explicitly by TLUSTY.

The equation of state, which essentially specifies a relation between mass density and electron density and/or the total number density, takes into account all the species.

We stress that the elemental abundances are specified in the standard input file for SYNSPEC. Recall that if nothing else is specified there, the abundances are taken as the values from the chosen *standard solar abundances* set – see § 4.1.

If this portion of the file `fort.2` is missing, all the species considered by SYNSPEC (i.e., all species unless specifically removed by setting MODE=0 in the standard input file – unit 5; see § 4.1) are taken into account in constructing the opacity table. If the additional part of `fort.2` is present, it contains a number of input records, one for each species that is going to be removed (or its abundance changed). Each line contains two numbers:

```
IATOM RELABN
```

where

IATOM – atomic number of the species to be removed/modified;

RELABN – a factor that multiplies the adopted standard solar abundance to that used for computing the opacity. The most usual option is RELABN=0. which removes the element completely.

For example, the file `fort.2` may look like

```
-----------------------------------------------------------------
11  3.0e3    1.5e4
1
7  1.0e-10  1.0e-6
100000  0  900. 110000.
'optab11_7hmc.dat'    0
  1   0.
 12   0.
 20   0.
-----------------------------------------------------------------
```

for constructing an opacity table analogous to that above, but removing hydrogen, magnesium, and calcium from the resulting opacity table.

4.9 Setting the dimensions of large arrays

Thanks to transporting the program to FORTRAN90/95, the array dimensions do not have to be hardwired, but can be dynamically allocated. Moreover, only arrays that are needed for a particular application have to be allocated. For instance, when computing a synthetic spectrum, all arrays relevant to opacity tables remain unallocated.

Some array dimensions follow directly from input parameters or files. For instance, `MDEPTH`, the maximum number of depth points, is set by the program based on the input model atmosphere, so the user does not have to specify it explicitly. The program also recognizes the kind of input model atmosphere, again without specifying it explicitly (except the case of a disk). Similarly, the number of frequencies in the opacity table is set by the input file `fort.2`, and the corresponding maximum dimension parameter `MFRTAB` is set to this value. All this is transparent to the user.

However, some dimensions cannot be so simply set. These are the actual numbers of spectral lines selected from the line list(s), the numbers of lines considered in one "set", and the internal number of frequencies when the opacity table is constructed. To allow for controlling the dimensions of corresponding arrays, which are typically quite large, but do depend on the king of selected application, we introduce an optional input file called `dimens` where these dimension can be specified. The corresponding arrays could thus have more realistic dimensions. Consequently, this may result in much more efficient and economic handling of the available computer memory. If the file `dimens` is not present, SYNSPEC takes the default values, which should be sufficient for most applications (although may waste memory in some cases).

At present, the following 5 parameters may be specified in the file `dimens`:

• `MLIN0`, the maximum number of lines selected from the atomic line list. The data for these selected lines are stored in several arrays containing necessary atomic data (e.g., `FREQ0, EXCL0, GF0` (storing line frequencies, excitation potential of the lower level, *gf*-values, respectively), and several others.

• The analogous quantity, `MLINM0`, is the number of lines selected from any one molecular line list (which, as discussed above, may be several).

It is very important to stress that if the current number of selected lines exceeds `MLIN0`, or `MLINM0`, *the program does not stop*. It computes the spectrum (or opacity table) till the wavelength of the last selected line (in fact, the spectrum is computed for a slightly smaller wavelength range, namely the last wavelength minus `CUTOFF`), and then the line list is being read again, and the parameters for the new set of lines is stored in the corresponding arrays starting with index 1. This was explained in more detail in § 3.5. When this occurs, the execution time of SYNSPEC is somewhat larger because of the repeated reading of the line list(s), but the experience shows that the computer time increase is very small. In any case, the saving of memory is certainly worth it.

This is particularly important for molecular line lists because there may be several of them (e.g. 20 in the example shown here, and some of them may contain a relatively small number of selected lines, so dimensioning all the arrays by the default dimension (10^6 in the present case), could lead to a gross waste of available computer memory.

• The parameters `MLIN` and `MLINM` represent the maximum number of lines selected for one "set" (typically about 1 Å wide). They specify dimensions of several 2-D arrays that contain quantities that depend both on frequency and depth (e.g. an opacity and emissivity). These numbers are typically small because not very many lines do contribute in such a small range of wavelengths, but one has to include lines whose centers are located beyond the given wavelength range, but they can still contribute to the opacity and emissivity through their wings.

Unlike the parameters `MLIN0` and `MLINM0`, program stops if the actual number of lines in a set exceeds `MLIN` (for atomic lines), or `MLINM` (for molecular lines). Therefore, the default values for these parameters were chosen to be quite conservative, and therefore safe.

• Finally, `MSFTAB` represents the maximum number of internal frequencies for computing an opacity table. We emphasize that the opacity table is first computed in a set of internal frequencies, set by SYNSPEC, and forming a sufficiently dense mesh so no lines are accidentally omitted, and then the opacities are interpolated to a set of frequencies (wavelengths), typically logarithmically equidistant, at which is the resulting opacity table stored.

The default values, and the selected values (if so specified) are displayed in the standard output, as shown below:

```
 DEFAULT DIMENSIONS OF LARGE ARRAYS
---------------------------------
MLIN0       1000000  (atomic lines stored)
```

```
MLIN          10000  (atomic lines in in one set)
MLINM0      1000000  (molecular lines per list stored)
MLINM         50000  (molecular lines in one set)
MSFTAB      6000000  (number of internal freq.points)


ACTUAL DIMENSIONS OF LARGE ARRAYS
---------------------------------
MLIN0       100000  (atomic lines stored)
MLIN          2000  (atomic lines in in one set)
MLINM0      200000  (molecular lines per list stored)
MLINM        50000  (molecular lines in one set)
MSFTAB     6000000  (number of internal freq.points)
```

The file `dimens` used in this case is:

```
MLIN0=100000
MLINM0=200000
MLIN=2000
MLINM=50000
```

Its structure is simple: each line contains a statement `parameter = value`. The names of the parameters may be in upper or lower case, and their order is arbitrary. If the parameter is missing in the file (e.g. `MSFTAB` in this case), its actual value is set to the default value.

5 Output files

The amount and structure of output depends on the basic mode of operation, set by keyword `IMODE`, and also on some other parameters - see below.

5.1 Ordinary spectrum synthesis

In the case of spectrum synthesis, `IMODE` = 0,1,2, the following output files are generated:

- Unit 6 - standard output file. It contains a log of calculations, important only in the case when something goes wrong.
- `fort.7` – synthetic spectrum. A simple table of wavelength [Å] versus flux, expressed as the first moment of the specific intensity (Eddington flux), H_λ, in erg cm^{-2}s^{-1}Å^{-1}.
- `fort.17` – An analogous table of a theoretical continuum flux.
- `fort.12` – Identification table. It contains a list of all lines that were selected by SYNSPEC to be taken into account, together with their basic

parameters and an indicator for their approximate strength. This information can be used for an identification of the lines in the predicted spectrum. It is generated only if `IPRIN`≥ 1 (or `IPRIN`≥ 2 in the opacity-table option);

- `fort.16` – A list of partial equivalent widths of SYNSPEC-generated elementary spectrum regions (typically around 1 Å wide). However, its importance is limited, and its use is outdated.

5.2 Opacity tables

In this case, set up by `IMODE` $= -3$ or -4, there are several output files generated. The most important one is obviously the opacity table itself. The name of the opacity table file is specified in the input file `fort.2`, see § 4.8. The file has the following structure (see also Paper IV, § 2.3):

1. A table of abundances of the first 92 elements, used when generating the table by SYNSPEC. There are two sets of abundances listed there, namely those used in the equation of state, and those used when computing detailed opacities[6].

2. A list of values for the keyword parameters IFMOL and TMOLIM used when generating the opacity table.

3. A list of additional opacity switches, referring to the opacities of H-, H_2^+, H_2^-, He-, the four CIA opacities, CH, and OH. A value of 1 means that the source was included when constructing the opacity table.

4. The numbers of frequencies, temperatures, and densities for the table.
   For irregular tables, the number of densities (now formal) is negative. Then there is an additional input record that for all temperatures specifies the number of density values.

5. A list of temperatures and densities in logarithmic scale.
   For irregular tables, there is a list of temperatures as usual, and then NTEMP *records with the individual values of the densities.*

6. A list of electron densities (again, in logarithmic scale) corresponding to the individual temperature-density pairs listed above.
   For irregular tables, the format is analogous to that of listing the density values.

7. For each frequency point, the value of the frequency, and the logarithms of the opacity (per gram) for all temperature-density pairs. The inner loop is over temperatures, and the outer loop over densities.
   For irregular tables, the order of loops is reversed.

[6] This has an important meaning when using the table in TLUSTY. In the case of using a full opacity table, that is with IOPTAB < 0, these two abundances are identical, while for a partial opacity table (IOPTAB > 0) the abundances of some species in the second column are set to 0, i.e. their opacities are removed from the opacity table, and are treated explicitly by TLUSTY.

Besides the TLUSTY-compatible opacity table, the program produces several additional output files that may be used for further inspection of the results, or in the case there are problems, but these files do not have to be stored.

fort.26 – continuum opacity, namely a table of continuum wavelengths, versus the opacity in the continuum, for each temperature/density pair.

fort.27 – all values of opacity on the internal wavelength grid for each temperature/density pair (i.e., before their interpolation to the universal wavelength grid). This is a rather big file, and it is produced only if the keyword parameter `IPRIN` (see § 4.2.1) is larger than 2.

fort.29 – an overview of the number of lines considered (atomic and molecular), and the computer time for each temperature density pair. The table is more or less self-explanatory. Briefly, the individual columns are: temperature index, density index, value of T [K], values of ρ [g cm^{-3}], electron density, number of atomic lines selected, number of molecular lines from the individual line lists (all in thousands), and the computing time spent to produce the opacities on the internal grid [s].

fort.63 – a binary version of the opacity table.

5.3 EOS-only option

The option is set up by the keyword parameter `IFEOS` > 0 (see § 4.3). It produces several output files with some interesting results of the chemical equilibrium (equation of state) solution. The run produces files `fort.51` – `fort.54`, which are more or less self-explanatory, and are discussed in detail in Paper IV, § 3.3. However, a usefulness of this option is now diminished because of a development of a new utility program CHEMEQ, which produces output files better suited for a graphical display. It The program is described in Appendix D.

5.4 Radiative accelerations

If the accelerations are computed by setting the keyword `IDIFF`=1, SYNSPEC produces an output file `fort.41`. Its structure is as follows:
The outer loop is over depth points; currently every second depth point is considered. For each depth there are currently 30 records, for the chemical species with atomic number 1 to 30 (H to Zn), each containing:
depth index, atomic number, and 6 values of radiative accelerations of the first 6 ionization stages, staring with neutrals.

6 Graphical user interface

Since the main output from SYNSPEC, the files `fort.7` (synthetic spectrum), and `fort.17` (theoretical continuum), are simple files with two columns – wavelength versus flux, plotting the results is trivial both in IDL as well as PYTHON. However, since the input to SYNSPEC is relatively simple and straightforward, we

have created two packages that serve as a wrapper to prepare the necessary input files, run the code, possibly perform rotational and instrumental convolutions, and plot the results.

6.1 Synplot – an IDL wrapper

SYNPLOT is an IDL wrapper to perform all the tasks outlined above, plus providing line identifications based on the output in `fort.12`. The program is described in a separate document. However, it has been developed to work with an older version of SYNSPEC. This original version of SYNPLOT, that works with SYNSPEC39, together with the the source code of SYNSPEC39 and the users's guide, can be downloaded from

`https://www.as.arizona.edu/~hubeny/pub/synplot2.1.tar.gz`

As mentioned in Paper IV, there is an upgrade of SYNPLOT, developed by Y.O., which is compatible with SYNSPEC54, and which works in IDL and GDL. The files are available at

`https://owncloud.iac.es/index.php/s/0CO8xQYTZXjJpxN`

6.2 Synple – a Python wrapper

Recently, one of us (C.A.P.) has developed a Python code, SYNPLE. It is to some extent analogous to SYNPLOT, although there are differences. In the present version, it does not allow to use all functionalities of SYNSPEC, but again, we plan to upgrade it soon.

Synple can take as input MARCS, Kurucz, Phoenix and Tlusty model atmospheres, and is designed to be useful to researchers that are not experts in stellar atmospheres and radiative transfer. It facilitates performing massive spectral synthesis calculations for large grids, parallelizing the calculations over multiple cores, using the python libraries `multiprocessing` or `ray`, and multiple processors, setting up scripts to run syn on distributed-memory computers.

The package can be downloaded from:

`https://github.com/callendeprieto/synple`

7 List of keyword parameters

Here we give the list of all keyword parameters in alphabetic order, including their default values, and the reference to the section where they are described in detail.

Keyword	*Default*	*Brief explanation*	*Ref*
ALAM0	0.	starting wavelength	4.3.1
ALAST	10000.	ending wavelength	4.3.1

ANG0	0.1	minimum angle cosine for spec.intensity	4.3.1
ERANGE	0.1	energy limits in NLTE levels association	4.3.3
CUTOF0	20.	line cutoff	4.3.1
GSSTD	1.e-5	param. for std. Stark broad. for mol.lines	4.3.2
GWSTD	1.e-7	param. for std. V.d.Waals broad. for mol.l.	4.3.2
HGLIM	1.e18	lower limit of $n_{\rm e}$ to use Gomez H line data	4.3.4
IATREF	1	index of "reference" atom	4.3.6
IBFAC	0	switch for input b-factors	4.3.6
ICHEMC	0	change of chem.abundances	4.3.1, 4.5
IDSTD	0	selection of std depth(s)	4.3.1
IEQBC	1	switch to use BC tables for equil.const.	4.3.2
IFEOS	0	switch to activate the "EOS-only" option	4.3.2
IFHE2	1	switch to consider He II as hydrogenic	4.3.1
IFLUX	0	switch for computing specific intensity	4.3.1
IFMOL	0 or 1[a]	switch to include molecules	4.3.2
IFREQ	1	radiative transfer solver	4.3.1
IHGOM	0	switch for Gomez Xenomorph H lines data	4.3.4
IHYDDK	0	switch for Koester new H Lyα, β data	4.3.4
IHYDPR	0	special hydrogen Stark profiles	4.3.1
IHE1PR	0	special He I Stark profiles	4.3.1
IHE2PR	0	special He II Stark profiles	4.3.1
ILPICK	1	switch to use L quant.num. for level assoc.	4.3.3
IMODE	0	basic mode of operation	4.3.1
INLIST	1	format of the atomic line list	4.3.1
INLTE	1	switch to force LTE for NLTE models	4.3.1
INMOD	1	kind of input model (obsolete)	4.3.1
INFBC	1	switch for BC tables for partition functions	4.3.2
IOPCH	1[b]	switch for CH continuum opacity	4.3.5
IOPH2H	1[b]	switch for H_2-H CIA opacity	4.3.5
IOPH2H2	1[b]	switch for H_2-H_2 CIA opacity	4.3.5
IOPH2HE	1[b]	switch for H_2-He CIA opacity	4.3.5
IOPH2M	1[b]	switch for H_2^- opacity	4.3.5
IOPH2P	1[b]	switch for H_2^+ opacity	4.3.5
IOPHEM	1	switch for He^- continuum opacity	4.3.5
IOPHMI	1	switch for H^- b-f and f-f opacity	4.3.5
IOPOH	1[b]	switch for OH continuum opacity	4.3.5
IPFEXO	1	switch for EXOMOL partition functions	4.3.2
IPPICK	1	switch to use parity for level assoc.	4.3.3
IPRIN	1	amount of printing	4.3.1
IRSCH2	1[c]	switch for H_2 Rayleigh scattering opacity	4.3.5
IRSCHE	1[c]	switch for He I Rayleigh scattering opacity	4.3.5
IRSCT	1[c]	switch for H Rayleigh scattering opacity	4.3.5
IRWTAB	1	switch to select a table for part.functs.	4.3.2
ISPICK	1	switch to use S quant.num. for level assoc.	4.3.3
IUNIM1	20	unit # for 1st molec. linelist (if no `fort.3`)	4.3.1

IUNIM2	21	unit # for 2nd molec. linelist (if no `fort.3`)	4.3.1
MOLTAB	1	switch to select a table for molecular EOS	4.3.2
NFREQC	2000	number of freqs. in continuum	4.3.1
NFREQS	120	number of frequency points in a "set"	4.3.5
NMLIST	0	number of molecular line lists	4.3.1
NMU	0	number of angles for spec.intensities	4.3.1
NUNALP	0	switch for quasi-mol. satellites of Lyα	4.3.1
NUNBAL	0	switch for quasi-mol. satellites of Hα	4.3.1
NUNBET	0	switch for quasi-mol. satellites of Lyβ	4.3.1
NUNGAM	0	switch for quasi-mol. satellites of Lyγ	4.3.1
NUNHE	0	switch for broadening of Lα by He	4.3.4
RELOP	1.e-4	line rejection parameter	4.3.1
SPACE	0.01	max. spacing of frequency points	4.3.1
TMOLIM	9000.	limiting temperature to consider molecules	4.3.2
VTB	2.	turbulent velocity	4.3.1

Notes:

(a): IFMOL set to 1 if $T_{\rm eff} < 9000$; otherwise IFMOL=0

(b): reset to 0 if IFMOL=0 or $T >$ TMOLIM

(c): reset to 0 if IMODE=-3 (opacity table)

Appendix A: Selection/rejection of lines from the line list

There are two ways of reducing the number of lines to deal with:

(i) The first one is to reduce the line list itself. This can be done either when the list is generated, as described in Appendix C, or to reduce the number of lines contained in an already existing list, thus creating a smaller list. This option is particularly useful for molecules with a huge number of lines, such as TiO or water, for which EXOMOL presents several 10^9 lines.

(ii) The second possibility is to take a line list as it is, and let SYNSPEC perform a selection of lines that may contribute, based on the real physical conditions in a studied atmosphere. This is obviously a safer option than the first one, but somewhat more time-consuming because SYNSPEC has to go through a bigger line list, and test many more lines for their strength. This procedure is controlled by two keyword parameters, `IDSTD` and `RELOP`, as explained in more detail in § 4.3.1/

A.1: Reducing the number of lines in the line list

The procedure was described in Paper IV, Appendix G. For convenience, we outline it here.

The opacity in the line center of a line between levels l and u of the molecular species "mol" is proportional to

$$\kappa(\nu_0) \propto (N_{\rm mol}/U_{\rm mol}) f_{lu} g_{lu} \exp(-E_l/kT)/\Delta\nu_D, \tag{2}$$

where N is the number density and U the partition function, f_{lu} the oscillator strength, g_l and E_l are the statistical weight and the energy of the lower level, respectively, and $\Delta\nu_D$ the Doppler width. The strength parameter of a line $l \leftrightarrow u$, defined as a logarithm of the line strength (without additive constants). can then be written as

$$s_{lu} = \log(N/U) + \log(gf) - 0.625 E_l/T + \log\lambda_0, \tag{3}$$

(we skipped subscript "mol" at N and U), and E_l is assumed to be expressed in cm^{-1}. Here, we have neglected the dependence of the Doppler width on the mass of the molecule, which is a reasonable approximation since the true thermal velocity is often smaller than the microturbulent velocity, so that $\Delta\nu_D \propto \nu_0$, and thus $1/\Delta\nu_D \propto \lambda_0$.

The strength parameter s still depends on the temperature (through the term E_l/T), as well as on density, through $\log(N_{\rm mol}/U_{\rm mol})$. In order to avoid the need of having tailored limited line lists for different temperatures and densities, we adopt the following procedure: We select a "characteristic" molecular species, for instance H_2O, at a characteristic wavelength, say 1μ. We then find, by extensive testing, what is the minimum strength parameter of a line of a characteristic species around this characteristic wavelength. In other words. it

is necessary to find what is s_{lu} of a line to contribute to the total opacity in an appreciable way, or, analogously, that the synthetic spectrum computed with and without this line being included differs by less than some chosen degree of accuracy, say 0.1%. We formulate a rejection criterion for a characteristic species in such a way that the line is rejected if

$$s \equiv \log gf - 0.625 E_l / T_{\rm char} + \log(\lambda/\lambda_{\rm char}) < s_{\rm reject}. \quad (4)$$

For instance, selecting $T_{\rm char} = 3000$ K and $\lambda_{\rm char} = 1\mu$, the value of $s_{\rm reject}$ that gave a reasonable accuracy of the predicted spectrum was about -9. Selecting lower values of $s_{\rm reject}$ yields a limited line list with more lines, but without yielding a significant difference in the predicted spectra.

To construct an analogous rejection criterion for a list that contains lines of several molecular species, we modify Eq. (4) to read

$$s \equiv \log(N/U)_{\rm max} - \log(N/U)_{\rm char} + \log gf - 0.625 E_l / T_{\rm char} + \log(\lambda/\lambda_{\rm char}) < s_{\rm reject}, \quad (5)$$

where $\log(N/U)_{\rm char}$ is the N/U value for the characteristic species at the characteristic temperature, and for the density where its concentration at this temperature is highest (typically $\rho \approx 10^{-6}$ g cm^{-3}). For water, and for the above specified T and ρ this value is $\log(N/U)^{\rm char} \approx 10$. In order to find $\log(N/U)_{\rm max}$, we first compute the values of N/U for a given molecule and for a number of temperatures and densities, and set the value of $T_{\rm char}$ to be the temperatures where N/U reaches the maximum, and take this N/U as $(N/U)^{\rm max}$. On this way, we can account for the fact that various molecules have various concentrations and various partition functions under different conditions, and that they are not spuriously disregarded by choosing the temperature and density where they are not formed in significant amounts.

It should be kept in mind that any a priori rejection procedure is only an approximate one, so the rejection parameter $s_{\rm reject}$ should be chosen conservatively, i.e. to be reasonably small. Our tests showed that the value $s_{\rm reject} \approx -8.5$ is a good and conservative choice for diatomic molecules. When computing synthetic spectra using a full list for diatomic molecules (with about 79.6 million lines), and a list reduced with $s_{\rm reject} = -8.5$ (with just 6 millions lines), the relative difference between the spectra for both a cool M-star model and a solar-type model, is around or less than 0.2%. The same conclusion can be drawn for TiO and water.

A.2: Short utility programs REDULIS and LIST2BIN

The program REDULIS performs a reduction of the line list, as mentioned above, and is described in detail in Appendix A.1. The code works for all three types of molecular line lists. The standard input contains either just one number, $s_{\rm reject}$ (see Appendix A.1) (which is mandatory), or three more quantities, $T_{\rm char}$, $\lambda_{\rm char}$, and $\log(N/U)_{\rm char}$ – see again Appendix A.1. If these values are not specified, the default values for these quantities, namely 3000, 10,000, and 10, respectively, are adopted.

The original list should be copied/linked to `fort.20`, and the output is on `fort.12`. The standard output contains information about how many lines of the individual molecular species are in the original list, and in the reduced one.

For completeness, we mention another extremely simple program, LIST2BIN. It simplty transforms the line list in the text (ASCII) format to a binary one. It is quite trivial, but is included for convenience. Since there are altogether 5 possible formats of a line list (2 for atomic, and 3 for molecular lists), the program automatically recognizes the format of the line list. The standard input is the text line list, and the output binary list is on file `fort.12`. It is advantageous to rename this file to have the same core name, with extension `.bin`.

Appendix B: Input of the NLTE departure coefficients

We first specify the definitions of the two types of the NLTE departure coefficients, usually called the b-=factors.

The "absolute" b-factors, defined as a ratio of NLTE to the (absolute) LTE populations,

$$b_i^* \equiv n_i/n_i^{**}, \tag{6}$$

where n_i is a NLTE population of level i, and n_i^{**} the *absolute* LTE population, obtained by solving the Saha-Boltzmann equations for all levels of all considered ionization stages of the element.

In contrast, the traditional b-factors (stored as before in the TLUSTY output file `fort.12`) are defined by

$$b_i \equiv n_i/n_i^* = n_i/[n_{\rm e} n_1^+ \Phi_i(T)], \tag{7}$$

where $n_{\rm e}$ is the electron density, n_1^+ the *actual* population of the ground state of the next ion, and $\Phi_i(T)$ is the Saha-Boltzmann factor – see Paper II, § 2.3, or Hubeny & Mihalas (2014, Chap. 9). This definition is useful from the point of view of understanding NLTE line formation from the theoretical point of view, while the definition we adopt here, Eq. (6) is useful from the point of view of abundance analysis, which in fact is one of the principal uses of SYNSPEC.

The traditional b-factors are stored in the TLUSTY output file `fort.12`, as in the previous versions, and the absolute b^*-factors are stored in the TLUSTY output file `fort.22`, which has exactly the same format as `fort.12`, and in fact also as `fort.7` – see Paper IV, § 2.10.2. To repeat, all the three files have the same header: the 1st line contains the number of depths and the number of physical parameters per depth; then several lines that contain the mass-depth coordinate (the column mass in g cm^{-2}), and then for each depth the temperature, electron density, mass density, total particle density (for models with molecules), and then for all explicit levels either the level populations (`fort.7`). or traditional b-factors (`fort.12`), or absolute b^*-factors (`fort.22`). Finally, we stress that the two latter files, `fort.12` and `fort.22`, are generated

only for NLTE models, because for LTE models all the b-factors are identically equal to unity.

For the use in SYNSPEC, the file containing the b-factors has to be copied/linked to the file `bfactors`; for more details refer to § 4.3.6.

Appendix C – Construction of isotope-averaged SYNSPEC-compatible line lists from EXOMOL data

Theoretical background is presented in Paper V, § 7.1. Here we describe the practical issues, such as the individual computer programs, and the way to use them efficiently.

C.1 Utility programs

It is advantageous to create a parent directory for EXOMOL data, say `./EXOMOL` with a number of subdirectories for the individual species, e.g. `./EXOMOL/AlH`, `./EXOMOL/AlO`, etc, as well as a subdirectory containing the utility programs, `./EXOMOL/xprog`.

There are four basic FORTRAN90/95 programs:

Program pfsum.f

This is a program that generates isotope-averaged partition function for a given molecular species, based on the EXOMOL data for the individual isotopologues. The relative abundance of the individual isotopologues is assumed to have solar values, which are given in the table stored in the file `isotps`, and also shown in Paper V. The EXOMOL partition functions are given as simple tables (partition function vers. temperature). In most cases, the values of the lowest and the highest temperature considered, T_0 and T_1, is the same for all isotopologues of the given molecular species, as is the of temperature step, but unfortunately not always. In such cases, the program does not extrapolate, but only takes values from the available range of temperatures. This does not lead to significant inaccuracies because the limited ranges of temperatures are usually presented for isotopologues with very small relative contributions to the total abundance of the given molecular species. If the ranges of temperatures are the same for all isotopologues, but the total numbers of temperatures are not (for instance, for CO, where there are 9000 temperatures the isotopologes containing ^{16}O, while only 3000 points for those containing ^{17}O and ^{18}O). In such cases, the program interpolates to the temperatures set composed of the largest number of temperatures.

The standard input to the program has $N + 1$ lines, with N being the total number of isotopologues. The first line is

iat1, iat2 - atomic numbers of the components

and then there is one line for each component, containing

iso1, iso2, filepf - where `iso1` and `iso2` are the nucleon numbers of the given atomic component, and `filepf` is a string containing the name of the EXOMOL partition function file.

For instance, for AlO, the input file, named `alo.5`, looks like:

```
13  8
27 16 '27Al-16O__ATP.pf'
27 17 '27Al-17O__ATP.pf'
27 18 '27Al-18O__ATP.pf'
```

The output, a table of isotope-averaged partition as a function of temperature, is the file `fort.10`.

Program list.f

Is a program to extract line data for a particular isotopologe of a molecule from the EXOMOL data and output them in the SYNSPEC format. It assumes that the total (isotope-averaged) partition function of the molecule is already given, either computed by a previous run of the program `pfsum`, or copied (linked) form some other source. Its name, however, should comply with the convention adopted here, namely containing just the chemical label, with the extension "`.pf`", e.g., `AlO.pf`.

Standard Input: There are four input lines organized as follows:

```
type   molname
iat1  iso1  iat2  iso2
core
wbeg wend gflim
```

where

type - a Kurucz-type label of a molecule, e.g 813.00 for AlO

molname - a character variable with the chemical label, e.g. 'AlO'

iat1, iso1 - atomic and nucleon numbers of the first atom, e.g 13, 27 for ^{27}Al

iat2, iso2 - the same for the second atom, e.g. 8 16 for ^{16}O

core - a character variable with the core name of the EXOMOL files, e.g. `'27Al-16O__ATP'`

wbeg - lowest wavelength [nm] for a line to be included in the list

wend - the same for the longest wavelength

gflim - $\log(gf)$ limit – lines with $\log(gf)$ below gflim are excluded.

For instance, the input file for the line list of $^{27}Al^{16}O$ (the file `cc1.5` of the example in the next section) is

```
813.00   'AlO'
13  27  8  16
'27Al-16O__ATP'
100 100000 -9.0
```

Output:

std.output - a short log file

fort.10 - the resulting line list in the SYNSPEC format, but in reverse wavelength order - from the long to short wavelengths. The accompanied short program `reverse.f` reverses it to the ordering needed by SYNSPEC , namely from short to long wavelengths.

Program reverse.f

This is a very simple program, with the standard input being the line list produced by `list.f`, and the standard output is the reversed list, in the SYNSPEC format.

Program merge.f

Again, a very simple program that merges two lists, the first one is being submitted as the standard input, and the second one as `fort.10`. The resulting merged file is stored in the output file `fort.11`. The program also generates a brief log file.

In the following example, it is assumed that these program are compiled, with the names of the executables `pfsum.exe`, `list.exe`, `reverse.exe`, and `merge.exe`.

C.2 Construction of line lists for individual species

Each subdirectory should contain two types of data:
(1) the files directly downloaded from the EXOMOL database;
(2) the input files to programs generating SYNSPEC-compatible line lists.

The files of the first type, are, for instance for the `./AlO` subdirectory, the following:

```
26Al-16O__ATP.pf
26Al-16O__ATP.states
26Al-16O__ATP.trans
```

and analogous triads `27Al-16O__ATP.*`, `27Al-17O__ATP.*`, and `27Al-18O__ATP.*`, that represent files obtained directly from the EXOMOL database. These files contain the partition functions (`*.pf`), energy level parameters (`*.states`), and the data for transitions (`*.trans`) , for the individual isotopologes.

The files of the second type have to be generated by the user. These are:
(1) the input file to the program `pfsum.f` to generate the isotope-averaged partition function (its example is showed in the previous section), or, alternatively, an already computed averaged partition function; and
(2) the script file `R1` to generate the line list, which in the case looks like this:

```
 ln -s -f ./xprog/isotops .
../xprog/list.exe <cc1.5 > cc1.log
../xprog/reverse.exe <fort.10 >cc1.list

../xprog/list.exe <cc2.5 > cc2.log
../xprog/reverse.exe <fort.10 >cc2.list

../xprog/list.exe <cc3.5 > cc3.log
../xprog/reverse.exe <fort.10 >cc3.list


cp cc1.list fort.10
../xprog/merge.exe <cc2.list >cc12.log
mv fort.11 cc12

cp cc12 fort.10
../xprog/merge.exe <cc3.list >cc13.log

# remove the isotope specific line lists (if not need for other purposes)
rm -f cc[1-3].list

mv fort.11 AlO.list
```

Here, the file `isotops`, located in the `./EXOMOL/xprog` directory, contains a table of the solar isotopic ratios for all important chemical elements. The utility programs `list, reverse`, and `merge` and the input data for them are described above.

C.3 Practical implementation

The package is distributed as a tar gzipped file `exomol.tar.gz`. Upon extracting, it generates directory `./exomol` with two subdirectories, `./exomol/xprog` and `./exomol/AlO`. The first one contains utility programs, as described before, as well as the file `isotops`. These four program need to be compiled, e.g.
`gfortran -o pfsum.exe pfsum.f`
and analogously for other programs.

Directory `AlO` represent an example of an individual directory for a given molecule, as described above. It contains the script file `R0` to generate the total partition function, the input file `alo.5` which is used by the script `R0` (via program `pfsum`). Then it contains script `R1` to generate the line list for AlO, and the input files `cc1.5, cc2.5` and `cc3.5` used by the script. The directory also contains file `AlO.pf-orig` for a check of a correct run of script `R0`.

Before running these scripts, the user has to download the corresponding `*.pf, *.states`, and `*.trans` files by visiting the EXOMOL website,
`https://www.exomol.com/data/molecules/`
and selecting the appropriate molecule, in this case AlO. In this particular case, one does not need to download files for 26Al-16O, albeit the data are present there, because the 27Al is the only isotope present in the Sun (see the table `isotops`). Once the files are downloaded, the user may run the script `R0` to get the total partition function, and then the script `R1` to generate the line list for AlO.

Analogous scripts and input files may be generated for other molecules.

Appendix D: Auxiliary program CHEMEQ

The program CHEMEQ is constructed from several slightly modified or reduced subroutines from SYNSPEC, designed to provide a fast solution of chemical equilibrium in the presence of molecular species. The program is thus useful for a determination of which molecular species are important at given atmospheric conditions (temperature and density), and therefore to help to decide which molecular line lists are needed for the potential spectrum synthesis using SYNSPEC. The program also provides an interesting insight into the question of what species are the individual atomic nuclei being sequestered in. Besides guiding the spectrum synthesis, the program may also have a pedagogical value.

The theoretical background was presented in Paper V, § 4.3 and 4.4.

D.1: Equilibrium constants and Partition functions

CHEMEQ uses the following default approach, analogous to that adopted in SYNSPEC with default settings of the keyword parameters, namely:
• for all molecules, the equilibrium constant is determined using the modified Tsuji-like table, as described in Paper V.
• If the parameter IEQBC is set to 1 (a default in CHEMEQ), the equilibrium constant is evaluated by an interpolation in the BC Table 7, for all 291 molecules for which BC provide data.

Although the partition functions do not directly enter in the chemical equilibrium evaluation when using pre-calculated values or fit formulae for equilibrium constants, the program CHEMEQ also presents partition functions for atoms, ions, and molecules, because these quantities are interesting from the point of view of spectrum synthesis. They are computed as in SYNSPEC with default keyword parameters.

D.2: Input and output

The input to CHEMEQ may be quite simple - one single line with the values of temperature T (in K), and the total particle number density (in cm^{-3}. If no other input files are present, the program assumes the solar composition (after Asplund et al. 2009), and the default evaluation of the equilibrium constants and partition functions. To change the default settings, two additional optional input files may be used.

In summary, the input files are as follows:

- standard input - mandatory: one line, contains values of T and N
- file `fort.56` - optional: values of chemical abundances that differ from the standard solar abundance set, analogously as in the standard SYNSPEC– see § 4.5. Briefly, the 1st line of the file specifies the number of changes in abundances, NCHANG.
  Then there are NCHANG lines, each containing two vales: the atomic number, and the abundance, ABUN. As in SYNSPEC,
  – a positive value of ABUN < 1 represents the absolute abundance (by number, relative to hydrogen);
  – a positive value of ABUN > 1 represents the abundance expressed as $12 - \log(N/N_H)$.
  – a negative number signifies that its absolute value represents the ratio of the abundance to the solar one.
- file `add.data` - optional: contains three parameters IEQBC, IPFBC, and IPFEXO that modify the default evaluation of the equilibrium constants (IEQBC) and/or partition functions (IPFBC and IPFEXO), exactly as in SYNSPEC. If not specified, the default values are:

  IEQBC = 1: i.e. the equilibrium constants are evaluated by interpolating from the BC tables for all molecular species considered by BC; the remaining species are treated using the original Tsuji data.

  IPFBC =1: analogous to previous, for partition functions. If any of the two above parameters is set to 0; the appropriate quantities are computed using the polynomial expansions.

  IPEXO=1: means that for some molecules the partition functions are interpolated from the tables taken from the EXOMOL database. At present, the following 32 molecular species are considered this way: AlO, C2, CH, CN, CO, CS, CaH, CaO, CrH, FeH, H2, HCl, HF, MgH, MgO, N2, NH, NO, NS, NaH, OH, PH, SH, SiH, SiO, SiS, TiH, TiO, VO, and 3 triatomic molecules H_2O, H_2S, CO_2.
  If IPFEXO is set to 0, the EXOMOL partition functions are not used.

There are two kinds of output, one that contains several tables for a visual inspection, and two output files that may be used for a graphical display of atomic, ionic, and molecular abundances, using a Python program `eos.py` which is a part of the current distribution package.

• In the standard output file, the first three tables are self-explanatory: table of atomic number densities and partition functions, and analogous tables for ions, and molecules.

Then there is a set of 15 tables for the following nuclei: H, C, N, O, F, Na, Mg, Al, Si, P, S, Ca, Ti, Cr, and Fe. Each table displays a list of species (molecular, as well as atomic) in which a given nucleus is sequestered. Each table shows an absolute number density (in cm^{-3}) of nuclei sequestered on a given species, as well as a number of nuclei in the given species divided by the total number of nuclei of the element (i.e., a relative distribution of nuclei over species). The individual species are ordered by decreasing number of sequestered nuclei, and only species with the relative distribution number larger than 10^{-15} are displayed.

• File `fort.78` - is a simple table of number densities of neutral and ionized atoms, and corresponding logarithms.

• File `fort.79` - is an analogous table of molecular number densities and their logarithms. The 1st column is a Tsuji-like index of the species.

D.3: Accompanied Python program EOS.PY

This is a short Python program to call program CHEMEQ, and plot resulting abundances of 92 neutral and ionized atoms and 503 molecular species, for s specified temperature, T, and one state parameter (total particle density, N_{tot}). A more detailed explanation is provided through comment lines in the code; here we demonstrate its use by simple examples:

1) To solve the atomic/molecular equilibrium for given parameters $T = 3000$ K, $N_{\mathrm{tot}} = 10^{14}$ cm^{-3}, and to produce initial plots:

```
>>> import eos
>>> eos.state(t=3000,ntot=1.e14)
```

2) To re-plot molecular abundances with a different marker shape and size:

```
>>> x,y,lab = eos.plotabund(mark='x',msize=4)
```

This will produce a rather cluttered plot, so it will be needed to enlarge it and zoom to a part of interest.

3) to overplot the current plot with previously computed file fort.79, called "prev.79", and without additional annotation:

```
>>> x,y,lab = eos.plotabund(file='prev.79',iover=1,annot=0)
```

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